

The Doubt Book

Fifteen-minute revision sheets

G10 — Essential Math and Math Plus

How to use this book

Each sheet is one reading activity, about fifteen minutes, meant for the start of a lesson. The sheets are numbered and listed in the index overleaf. You do not have to work through them in order, but within a strand each sheet assumes the one before it.

Work down the page in order. When you reach a **Pause**, stop and write something on the ruled lines before you read on — even a guess. The explanation that follows will settle it. The pauses are the point of the sheet; skipping them turns a thinking activity into a reading activity.

The last question on every sheet is different. It has no explanation after it, only a box to work in. Try it properly before you turn to **Check after you try** at the foot of the sheet.

Where something is left blank, that is the space meant for you. If you need more room, use the facing page.

Index

Trigonometry

- 1** Right-angled triangles — choosing a ratio
- 2** The area of any triangle — $A = \frac{1}{2}ab \sin C$
- 3** The sine rule — finding a side
- 4** The sine rule — two possible angles
- 5** How many triangles? The height test
- 6** The cosine rule — finding a side
- 7** The cosine rule — finding an angle, including obtuse
- 8** Bearings
- 9** 3D trigonometry — finding the right triangle inside a solid
- 10** The angle between a line and a plane

Proof

- 11** What makes an argument a proof
- 12** One letter, or two?
- 13** Difference of two squares as a proof tool
- 14** Place value — two-digit numbers and their reverse
- 15** Visual proof — dissecting a square
- 16** Visual proof — $a^2 - b^2 = (a + b)(a - b)$
- 17** Visual proof — Pythagoras

Exponents

- 18** Index laws, and zero, negative and fractional indices
- 19** Common base, then equate indices
- 20** When the question wants an expression, not x
- 21** Quadratics in disguise
- 22** Simultaneous exponential equations
- 23** Substituting one index relation into another
- 24** When the bases will not match — take logarithms

Logarithms

- 25** What a logarithm means, and exact values
- 26** The laws — splitting
- 27** Solving log equations, and rejecting roots
- 28** Change of base
- 29** Equations with mixed bases
- 30** Substitution — quadratics in logs and in e^x

Sheet 1 Right-angled triangles — choosing a ratio

15 minute revision · Calculator in degree mode

Your aim. You will learn to label a right-angled triangle relative to an angle you name, and choose the ratio that uses the two sides you already have.

Before you start. The hypotenuse is the side opposite the right angle, and it is always the longest side. Check your calculator shows DEG before you start.

The central idea

Every right-angled triangle question begins with a decision rather than a calculation: **which angle are you working with?** The words *opposite* and *adjacent* mean nothing until you have named an angle, because they describe the sides in relation to it.

Once you have named the angle, the three sides take fixed roles:

The **hypotenuse** lies opposite the right angle. It is the longest side, and it stays the hypotenuse whichever angle you name. The **opposite** side faces your named angle from across the triangle. The **adjacent** side is the one left over, running from your named angle to the right angle.

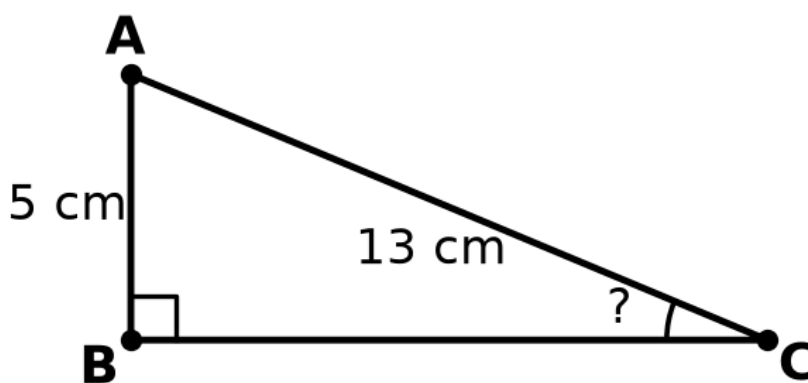
Each ratio uses a different pair of those three:

$$\sin x = \text{opposite} \div \text{hypotenuse}$$

$$\cos x = \text{adjacent} \div \text{hypotenuse}$$

$$\tan x = \text{opposite} \div \text{adjacent}$$

So the method chooses itself. Label the sides for your angle, look at which two lengths you have, and take the ratio that uses exactly those two.



Triangle ABC, right-angled at B, with AB = 5 cm and AC = 13 cm. Not to scale.

Pause 1 meaning

In triangle ABC above, which side is the hypotenuse, and which side is opposite $\angle ACB$?

Follow one example

We want $\angle ACB$, so $\angle ACB$ is the angle we name. AC lies opposite the right angle at B, so AC = 13 cm is the hypotenuse. AB = 5 cm faces $\angle ACB$ from across the triangle, so it is the opposite side. We have the opposite and the hypotenuse and we want the angle.

That pair is the sine ratio, so sine is the method.

$$\sin(\angle ACB) = \text{opposite} \div \text{hypotenuse} = 5 \div 13$$

Pause 2 *choosing the next step*

You now have $\sin(\angle ACB) = 5 \div 13$, but the question asks for the angle itself. What do you press on the calculator, and why is it not the sin button?

Dividing gives $\sin(\angle ACB) = 0.384615\dots$, which tells you the ratio but not the angle. To go backwards from a ratio to the angle that produces it, use the inverse function $\sin^{-1}$:

$$\angle ACB = \sin^{-1}(0.384615\dots) = 22.6^\circ \text{ (to 1 d.p.)}$$

Keep the full 0.384615... in the calculator rather than retyping 0.38. Round once, at the end.

The condition that matters

Nothing about the triangle changed in that example — only which angle we named. Name a different angle and every label except the hypotenuse changes with it.

Pause 3 *choosing the next step*

Find $\angle BAC$ instead, using the same two given lengths, 5 cm and 13 cm. Which side is now the opposite, which is the adjacent, and which ratio will you use?

For $\angle BAC$, the side AB = 5 cm now runs from the named angle to the right angle, so it has become the **adjacent** side. The hypotenuse is still AC = 13 cm. Adjacent with hypotenuse is the cosine ratio: $\cos(\angle BAC) = 5 \div 13$, giving $\angle BAC = 67.4^\circ$. The two angles add to 90.0° , which is the check — in a right-angled triangle the other two angles must.

Pause 4 *validity*

A student's working finishes with $\cos x = 1.4$. Without seeing the rest of their page, how do you know something has gone wrong?

Both sine and cosine divide a shorter side by the hypotenuse, and the hypotenuse is the longest side, so neither ratio can ever exceed 1. A value above 1 means the sides have been divided the wrong way round, or the wrong ratio was chosen. Tangent is the exception — it has no upper limit, because neither of its sides is the hypotenuse.

Try a fresh question

5. In triangle PQR the right angle is at Q. $PQ = 9$ cm and $\angle PRQ = 34^\circ$. Find the length of PR and the length of QR, each to 1 decimal place. Keep full accuracy until the end, then check your two answers against Pythagoras.

Remember. Name the angle first, label the sides against it, choose the ratio that uses the two lengths you have, and use $\sin^{-1}$, $\cos^{-1}$ or $\tan^{-1}$ only when the unknown is an angle. Check the calculator is in degrees, and that no sine or cosine you find is greater than 1.

Check after you try

1. $AC = 13$ cm is the hypotenuse, because it lies opposite the right angle at B; $AB = 5$ cm is opposite $\angle ACB$.
2. $\sin^{-1}$. The sin button turns an angle into a ratio; you have the ratio and want the angle, so you need the inverse.
3. $AB = 5$ cm is now the adjacent side and $AC = 13$ cm is still the hypotenuse, so use cosine; $\angle BAC = 67.4^\circ$.
4. A cosine cannot exceed 1, because the adjacent side is never longer than the hypotenuse.
5. $\angle PRQ = 34^\circ$ with $PQ = 9$ cm opposite it and PR the hypotenuse, so $\sin 34^\circ = 9 \div PR$, giving $PR = 16.1$ cm. QR is adjacent to the 34° , so $\tan 34^\circ = 9 \div QR$, giving $QR = 13.3$ cm. Check with the unrounded values: $9^2 + 13.343^2 = 259.0$ and $16.095^2 = 259.0$, so the two answers are consistent. Using the rounded 13.3 and 16.1 instead gives 257.9 against 259.2, which is why the check must be done before rounding.

Sheet 2 The area of any triangle — $A = \frac{1}{2}ab \sin C$

15 minute revision · Calculator in degree mode

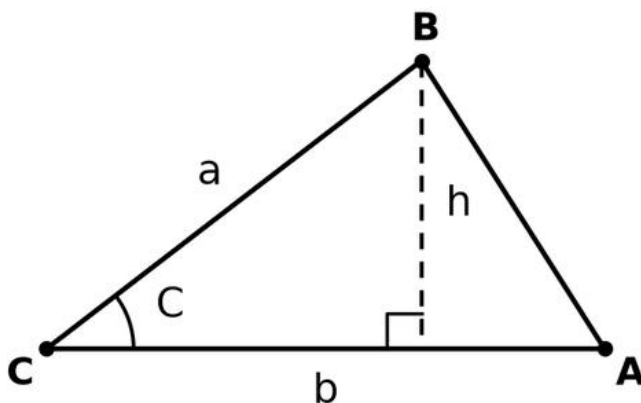
Your aim. You will learn to derive the formula $A = \frac{1}{2}ab \sin C$ from the area you already know, and recognise when you are allowed to use it.

Before you start. The area of a triangle is $\frac{1}{2} \times \text{base} \times \text{height}$, where the height is measured perpendicular to that base. In a right-angled triangle, $\sin$ of an angle is the opposite side divided by the hypotenuse.

The central idea

You already know that a triangle's area is $\frac{1}{2} \times \text{base} \times \text{height}$. The difficulty in most questions is that nobody gives you the height — you are given two sides and the angle between them instead. This formula closes that gap by building the height out of what you were given.

Take triangle ABC and call the two sides that meet at C by their usual names: a is the side from C to B, and b is the side from C to A. Use b as the base. Drop a perpendicular from B down onto b and call its length h .



The perpendicular h dropped from B onto the base b . Not to scale.

That perpendicular creates a right-angled triangle in which the angle at C is C , the side opposite C is h , and the hypotenuse is a . So

$\sin C = h \div a$, which rearranges to $h = a \sin C$.

Now substitute that height into the area you already know:

$$A = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times b \times h = \frac{1}{2} \times b \times a \sin C = \frac{1}{2}ab \sin C$$

The formula is not a new fact. It is $\frac{1}{2} \times \text{base} \times \text{height}$ with the height written in terms of the angle.

Pause 1 *meaning*

Look at where C sits in relation to a and b . What has to be true about the angle before you are allowed to write $\frac{1}{2}ab \sin C$?

C must be the angle **between** the two sides a and b — the included angle. That is not an extra rule to memorise; it is forced by the derivation, because h was built from the right angle at the foot of the perpendicular and the angle at C.

Follow one example

In triangle ABC, $a = 7$ cm, $b = 9$ cm and the angle between them is $C = 52^\circ$. Find the area.

The two sides and the angle between them are exactly what the formula needs, so no other working is required first.

$$A = \frac{1}{2} \times 7 \times 9 \times \sin 52^\circ$$

Pause 2 *choosing the next step*

Before pressing anything: is $\sin 52^\circ$ a length, an angle, or just a number? What are the units of the final answer?

$\sin 52^\circ$ is a ratio of two lengths, so it is a plain number with no units — here about 0.7880. The units come only from the two lengths, $\text{cm} \times \text{cm}$, so the area is in cm^2 .

$$A = \frac{1}{2} \times 7 \times 9 \times 0.78801\dots = 24.8 \text{ cm}^2 \text{ (to 1 d.p.)}$$

The condition that matters

Pause 3 *choosing the next step*

A different triangle has sides 7 cm and 9 cm, and an angle of 52° which lies **opposite** the 7 cm side rather than between the two sides. Can you still write the area as $\frac{1}{2} \times 7 \times 9 \times \sin 52^\circ$? Explain.

No. The formula needs the angle enclosed by the two sides you are multiplying. With the 52° opposite one of them you would first have to find the angle between them, which is a sine rule problem — Sheets 3 and 4. Writing $\frac{1}{2}ab \sin C$ with the wrong angle is the most common way to lose every mark on an area question.

Pause 4 *validity*

Work out the area when the included angle is 128° instead of 52° , keeping $a = 7$ and $b = 9$. Compare it with the answer above. What do you notice, and can you explain it?

You get 24.8 cm^2 again, because $\sin 128^\circ = \sin 52^\circ$. Any angle and its supplement have the same sine, so two genuinely different triangles — one with a sharp corner at C, one with a blunt one — can enclose exactly the same area. The formula does not break for an obtuse angle; the perpendicular simply falls outside the triangle, and $h = a \sin C$ still holds. That equality of sines is worth remembering: it is the whole subject of Sheet 4.

Try a fresh question

5. In triangle PQR, $PQ = 11 \text{ cm}$, $QR = 6 \text{ cm}$ and $\angle PQR = 73^\circ$. Find the area of the triangle to 1 decimal place. Then state, without a second calculation, the area of a triangle with the same two sides but an included angle of 107° , and say why.

Remember. $A = \frac{1}{2}ab \sin C$ multiplies two sides by the sine of the angle **between** them. It is $\frac{1}{2} \times \text{base} \times \text{height}$ with $h = a \sin C$ substituted in. If the given angle is not the included angle, you must find the included angle before you may use it.

Check after you try

- 1. C must be the angle between sides a and b — the included angle — because the derivation builds h from that angle.
- 2. A number, with no units: it is one length divided by another. The $\text{cm} \times \text{cm}$ gives the area its cm^2 units.
- 3. No. The formula requires the angle enclosed by the two sides being multiplied; here the 52° is opposite one of them, so the included angle must be found first.
- 4. 24.8 cm^2 , the same, because $\sin 128^\circ = \sin 52^\circ$. Supplementary angles have equal sines.
- 5. $A = \frac{1}{2} \times 11 \times 6 \times \sin 73^\circ = 31.6 \text{ cm}^2$. With 107° the area is also 31.6 cm^2 , because $107^\circ = 180^\circ - 73^\circ$ and so has the same sine.

Sheet 3 The sine rule — finding a side

15 minute revision · Calculator in degree mode

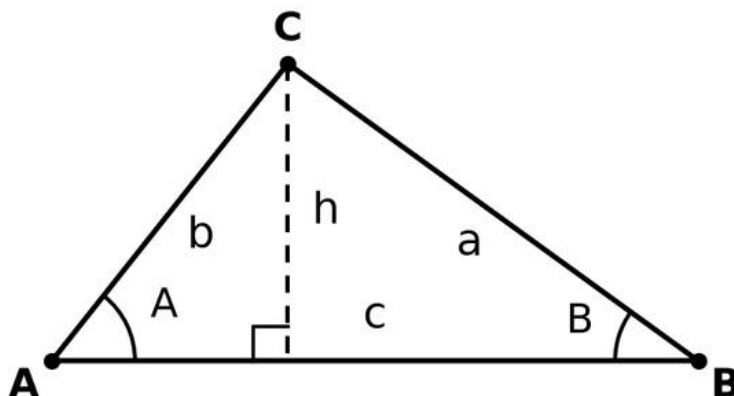
Your aim. You will learn to derive the sine rule from a single perpendicular height, and use it to find a missing side.

Before you start. In a right-angled triangle, $\sin$ of an angle is the opposite side over the hypotenuse. Sides are named with the lower-case letter matching the angle opposite them, so side a lies opposite angle A .

The central idea

The area formula on Sheet 2 came from dropping one perpendicular. Drop the same perpendicular in a triangle with no right angle and you get a second, more useful result.

Take triangle ABC with the side $c = AB$ along the bottom, and drop a perpendicular of length h from C onto AB . You have now split the triangle into two right-angled triangles that **share the same h** .



One perpendicular h from C , shared by two right-angled triangles. Not to scale.

In the left-hand triangle the angle is A , the side opposite it is h , and the hypotenuse is b . In the right-hand triangle the angle is B , the side opposite it is h , and the hypotenuse is a . So

$$\sin A = h \div b \text{ and } \sin B = h \div a,$$

which rearrange to $h = b \sin A$ and $h = a \sin B$. These are two descriptions of one and the same length, so they must be equal:

$$b \sin A = a \sin B$$

Divide both sides by $\sin A \sin B$ and the sines swap sides:

$$a \div \sin A = b \div \sin B = c \div \sin C$$

That is the sine rule. Each fraction pairs a side with the angle **directly opposite it**. The third fraction follows by dropping the perpendicular from a different vertex.

Pause 1 *meaning*

Side b is paired with angle B in the rule. In triangle ABC , where does angle B sit in relation to side b ?

Angle B is opposite side b — they face each other across the triangle and are never adjacent. Matching a side to the wrong angle is the only real way to go wrong here, so mark the opposite pairs on the diagram before you write anything.

Follow one example

In triangle ABC, $A = 47^\circ$, $B = 63^\circ$ and $a = 8$ cm. Find b.

Check first that the rule applies: you need one **complete opposite pair** plus one more piece. Here A and a are a complete pair, and B is the extra piece, so the sine rule is the method. No right angle is needed.

Because the unknown is a **side**, write each fraction side-over-sine, so that b ends up on top and one multiplication finishes it:

$$b \div \sin 63^\circ = 8 \div \sin 47^\circ$$

Pause 2 *choosing the next step*

Which single operation turns that line into a value for b, and why is it allowed?

Multiply both sides by $\sin 63^\circ$. Both sides of an equation may be multiplied by the same non-zero number, and $\sin 63^\circ$ is not zero, so this is legitimate:

$$b = 8 \times \sin 63^\circ \div \sin 47^\circ = 9.75 \text{ cm (to 3 s.f.)}$$

Sanity check: b should be longer than a, because 63° is a bigger angle than 47° and the bigger angle always faces the longer side. It is.

The condition that matters

Pause 3 *choosing the next step*

Suppose you are given $A = 47^\circ$, $B = 63^\circ$ and side **c** = 8 cm, and asked for b. You no longer have a complete opposite pair. What is your first move?

Find the third angle: $C = 180^\circ - 47^\circ - 63^\circ = 70^\circ$. That completes the pair C with c, and the sine rule becomes available. Whenever two angles are given, the third is free — take it before deciding the rule cannot be used.

Pause 4 *validity*

A student writes $\sin 47^\circ \approx 0.73$, then computes $8 \times 0.891 \div 0.73$. Their answer is 9.76 cm rather than 9.75 cm. Which is right, and what is the general rule?

9.75 cm. Rounding $\sin 47^\circ$ from 0.731354... to 0.73 changes the answer in the second decimal place, and in a longer calculation the error grows. Carry full calculator accuracy all the way through and round once, at the very end.

Try a fresh question

5. In triangle XYZ, $\angle X = 38^\circ$, $\angle Z = 25^\circ$ and $XY = 14$ cm. Find the length of YZ to 3 significant figures. Name the opposite pair you used.

Remember. The sine rule pairs each side with the angle opposite it. Use it when you have one complete opposite pair plus one more side or angle. Finding a side — write it side-over-sine. If no pair is complete, look for the third angle first.

Check after you try

1. Opposite it. Side b and angle B face each other across the triangle.
2. Multiply both sides by $\sin 63^\circ$, which is allowed because it is a non-zero number; $b = 9.75$ cm.
3. Find the third angle, $C = 180^\circ - 47^\circ - 63^\circ = 70^\circ$, which completes the pair C with c.
4. 9.75 cm is right. Rounding an intermediate value shifts the answer; keep full accuracy and round only at the end.
5. $XY = 14$ cm is opposite $\angle Z = 25^\circ$, so that is the complete pair. YZ is opposite $\angle X = 38^\circ$, so $YZ \div \sin 38^\circ = 14 \div \sin 25^\circ$, giving $YZ = 20.4$ cm. It is longer than XY, as it should be, since $38^\circ > 25^\circ$.

Sheet 4 The sine rule — two possible angles

15 minute revision · Calculator in degree mode

Your aim. You will learn to find every possible angle from a sine, and explain which of the answers actually fit a triangle.

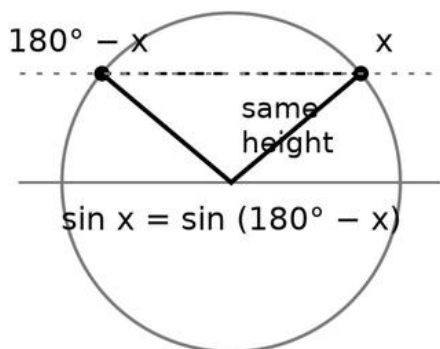
Before you start. The sine rule pairs each side with the angle opposite it. Side a is opposite angle A , side b is opposite B , side c is opposite C . You need one complete opposite pair plus one more side or angle; you do not need a right angle.

The central idea

Sheet 3 used the sine rule to find a **side**, and there was one answer. Turn the rule over to find an **angle** and something changes: the calculator gives you one answer, but the triangle may allow two.

The reason is that sine does not distinguish between an angle and its supplement. Picture a point moving round a circle of radius 1. The sine of the angle is the **height** of that point above the centre line. An acute angle and its supplementary obtuse angle put the point at two different places at exactly the same height, so

$\sin x = \sin(180^\circ - x)$ for every angle x .



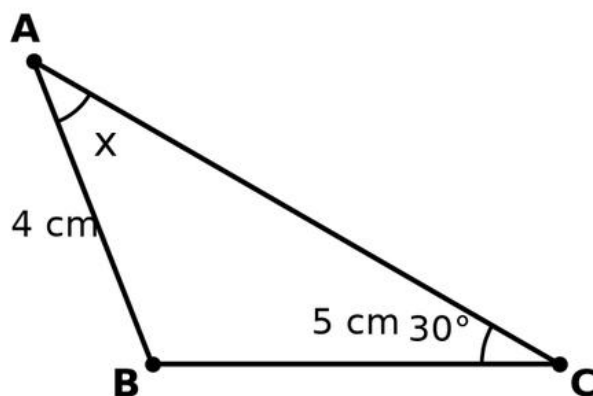
Two angles, one height. The acute angle x and the obtuse angle $180^\circ - x$ have equal sines.

Your calculator must return a single number, so $\sin^{-1}$ always gives you the acute one. The obtuse partner is real, and in a triangle it is often a genuine second answer. **It is your job, not the calculator's, to produce it and then test it.**

Follow one example

In triangle ABC , $AB = 4$ cm, $BC = 5$ cm and $\angle ACB = 30^\circ$. Find x , the angle BAC .

The side opposite the 30° angle is $AB = 4$ cm, so 4 and $\sin 30^\circ$ form the complete pair.



One triangle that fits the measurements. Not to scale.

Pause 1 *meaning*

Which side is opposite x , and how long is it?

$BC = 5$ cm lies opposite x . Because the unknown is an **angle**, write each fraction sine-over-side so that $\sin x$ finishes on top:

$$\sin x \div 5 = \sin 30^\circ \div 4 \text{ so } \sin x = 5 \times \sin 30^\circ \div 4 = 0.625$$

Pause 2 *validity*

The inverse sine button gives $x \approx 38.7^\circ$. Could a different interior angle of a triangle have the same sine? Write down a candidate.

Why another angle is possible

Yes — the supplement. Since $\sin x = \sin(180^\circ - x)$, the two candidates here are

$$x = 38.7^\circ \text{ or } x = 180^\circ - 38.7^\circ = 141.3^\circ$$

They are only **candidates** until you check that a triangle can actually contain them. The test is the third angle: $B = 180^\circ - 30^\circ - x$ must come out greater than 0° . If the other two angles already use up 180° , there is no room left for a third.

Candidate for x	Remaining angle B	Fits the given information
38.7°	111.3°	Yes
141.3°	8.7°	Yes

Both survive, so **both triangles genuinely exist** on these measurements. A sketch cannot choose between them. Only extra wording in the question — something like "angle A is obtuse" — would select one.

Pause 3 *choosing the next step*

In a different problem the candidates for A are again 38.7° and 141.3° , but this time $C = 80^\circ$. Which candidate must you reject, and why?

Reject 141.3° , because $141.3^\circ + 80^\circ = 221.3^\circ$, already more than 180° , leaving a negative angle for B. So the second candidate is not automatically a second answer — sometimes the angle sum kills it, and then there is exactly one triangle.

Pause 4 *validity*

A calculation gives $\sin A = 1.08$. How many triangles are possible? Explain briefly.

None. A sine is a ratio that can never exceed 1, so no angle has a sine of 1.08 and no triangle fits those measurements. Altogether there can be **zero, one or two** triangles: a sine above 1 gives none; a sine of exactly 1 gives only the candidate 90° ; and a sine strictly between 0 and 1 gives two candidates, each of which must be tested against the angle sum.

Try a fresh question

5. In triangle ABC, $AB = 6$ cm, $BC = 8$ cm and $C = 35^\circ$. Find every possible value of A and its matching value of B. Give your answers to 1 decimal place, keeping full calculator accuracy until the end.

Remember. Match the opposite pairs, find $\sin x$, then write **both** candidates x and $180^\circ - x$. Test each against the angle sum: the third angle must be greater than 0° . Apply any extra condition the question states.

Check after you try

1. BC, 5 cm.
2. 141.3° .
3. Reject 141.3° , because $141.3^\circ + 80^\circ$ is already more than 180° .
4. Zero; a real sine cannot exceed 1.
5. $\sin A = 8 \times \sin 35^\circ \div 6 = 0.7648$, so $A = 49.9^\circ$ or 130.1° . The matching pairs are $(A, B) = (49.9^\circ, 95.1^\circ)$ or $(130.1^\circ, 14.9^\circ)$. Both remaining angles are positive, so both triangles exist.

Sheet 5 How many triangles? The height test

15 minute revision · Calculator in degree mode

Your aim. You will learn to decide from a picture, before any sine rule working, whether two sides and a non-included angle give no triangle, one triangle or two.

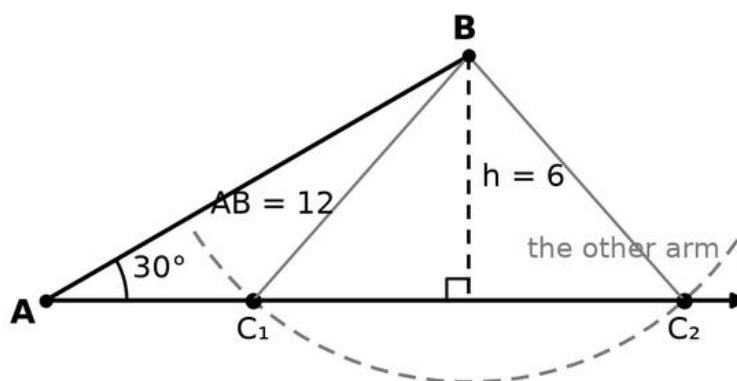
Before you start. Sheet 4 — a sine gives two candidate angles, and each must be tested. The perpendicular distance from a point to a line is the shortest distance from that point to it.

The central idea

Sheet 4 settled the question algebraically: work out the sine, take both candidates, test each one. This sheet answers the same question **geometrically**, before any calculation, and it is the method the notes use.

Suppose you are told one angle, the side beside it, and the side opposite it. Fix what you are certain of: put angle A at a point, and draw the two arms of that angle. Mark B along one arm at the given distance AB. Now the third vertex C must lie somewhere on the other arm, at the given distance BC from B.

So take a compass, put the point on B, open it to BC, and swing an arc. **Every place where that arc meets the other arm is a triangle.** The arc can miss the line, touch it once, or cut it twice — which is exactly zero, one or two triangles.



Swinging an arc of radius BC from B. The dashed perpendicular h is the shortest reach. Not to scale.

Two lengths decide everything. The first is **h, the perpendicular distance from B to the line AC** — the shortest reach the arc has. The second is **AB itself**.

Because the perpendicular from B makes a right-angled triangle with the angle A and hypotenuse AB,

$$h = AB \times \sin A.$$

Pause 1 meaning

The arc is centred on B with radius BC. What must be true about BC for the arc to reach the line AC at all?

BC must be at least h . If BC is shorter than the shortest distance from B to the line, the arc never reaches it and no triangle can close.

Follow one example

In triangle ABC, $AB = 12$ cm and $\angle BAC = 30^\circ$. Find the values of BC giving (a) no triangle, (b) exactly one triangle, (c) two triangles.

First the critical height:

$$h = 12 \times \sin 30^\circ = 6 \text{ cm}$$

Pause 2 *choosing the next step*

You now have $h = 6$ and $AB = 12$. Which of these does BC have to be compared with, and what do you expect to happen when BC is between them?

Compare BC with both. Reading the arc from short to long:

$BC < 6$ — the arc falls short of the line. **No triangle.**

$BC = 6$ — the arc just touches the line, at the foot of the perpendicular, in one place. **One triangle**, right-angled at C.

$6 < BC < 12$ — the arc crosses the line twice, and both crossings lie on the arm of the angle. **Two triangles**, and they are the two candidates of Sheet 4.

$BC \geq 12$ — the arc still cuts the line twice, but one crossing has slipped behind A onto the wrong side. That point does not lie on the arm of the 30° angle, so it does not give a triangle with the angle we were told. **One triangle.**

The condition that matters

Pause 3 *validity*

The $BC \geq 12$ case is the one students lose. The arc plainly cuts the line twice, so why is the answer one triangle and not two?

Because an angle has arms, not a full line. Only crossings on the correct arm count. When $BC = AB = 12$ the second crossing lands exactly on A, which collapses the triangle altogether; when $BC > 12$ it lands beyond A, where the angle at A would no longer be 30° .

Pause 4 *validity*

Test the geometry against Sheet 4. Put $BC = 4$ into the sine rule, $4 \div \sin 30^\circ = 12 \div \sin C$, and find $\sin C$. Does the algebra agree with the picture?

$\sin C = 12 \times \sin 30^\circ \div 4 = 1.5$, which is impossible, so there is no triangle — exactly what the arc showed. The two methods agree everywhere. $BC = 6$ gives $\sin C = 1$ and the single right angle; $BC = 8$ gives $\sin C = 0.75$ and two candidates that both survive the angle sum; $BC = 12$ gives $\sin C = 0.5$ and candidates 30° and 150° , the second leaving $B = 0^\circ$ and so rejected. The picture and the algebra are one method in two costumes, and the picture is faster.

Try a fresh question

5. In triangle ABC, $AB = 10$ cm and $\angle BAC = 40^\circ$. Find the values of BC for which there is (a) no triangle, (b) exactly one triangle, (c) two triangles. Give lengths to 2 decimal places.

Remember. Fix the angle and the side beside it, then swing the opposite side as an arc. Compare it with $h = AB \sin A$ and with AB itself: shorter than h , none; equal to h , one; between h and AB , two; equal to or longer than AB , one.

Check after you try

1. BC must be at least h . Shorter than the perpendicular distance and the arc cannot reach the line.
2. Compare with both $h = 6$ and $AB = 12$; between them the arc cuts the arm twice, giving two triangles.
3. Only crossings on the arm of the angle count. At $BC = 12$ the second crossing falls on A itself, and beyond that it falls behind A , where the angle would not be 30° .
4. $\sin C = 1.5$, which is impossible — no triangle, agreeing with the arc that falls short.
5. $h = 10 \times \sin 40^\circ = 6.43$ cm. (a) No triangle for $BC < 6.43$ cm. (b) Exactly one for $BC = 6.43$ cm or for $BC \geq 10$ cm. (c) Two for $6.43 \text{ cm} < BC < 10$ cm.

Sheet 6 The cosine rule — finding a side

15 minute revision · Calculator in degree mode

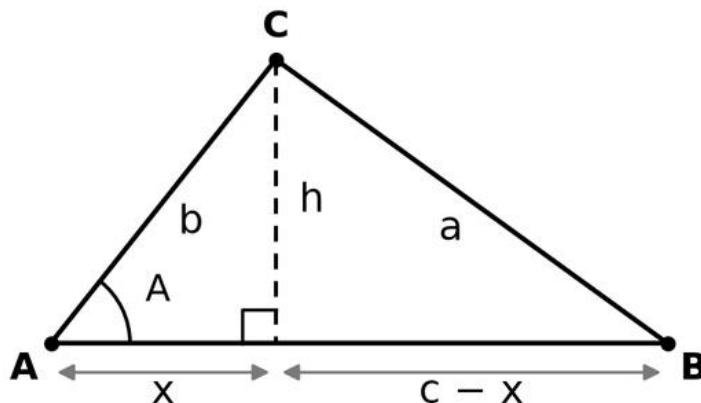
Your aim. You will learn to derive $a^2 = b^2 + c^2 - 2bc \cos A$, and use it when the sine rule cannot start.

Before you start. Pythagoras' theorem in a right-angled triangle. In a right-angled triangle, $\cos$ of an angle is the adjacent side over the hypotenuse. Expanding $(c - x)^2 = c^2 - 2cx + x^2$.

The central idea

The sine rule needs a complete opposite pair — a side and the angle facing it. Give someone two sides and the angle **between** them and the sine rule cannot even start, because the angle you know faces the side you do not. The cosine rule is built for exactly that situation.

It comes from the same move as Sheets 2 and 3: drop one perpendicular. Put $c = AB$ along the bottom, drop h from C onto AB , and let the foot of the perpendicular cut AB into two pieces, x next to A and $c - x$ next to B .



The base c split into x and $c - x$ by the perpendicular h . Not to scale.

In the left-hand right-angled triangle, x is adjacent to angle A and b is the hypotenuse, so $\cos A = x \div b$, which gives $x = b \cos A$. Hold that result back for now — it goes in at the very end.

Apply Pythagoras in each right-angled triangle:

left: $x^2 + h^2 = b^2$ so $h^2 = b^2 - x^2$ right: $(c - x)^2 + h^2 = a^2$

Substitute the first into the second to remove h , which is the whole purpose of h — it gets us between the two triangles and then leaves:

$$(c - x)^2 + b^2 - x^2 = a^2 \quad c^2 - 2cx + x^2 + b^2 - x^2 = a^2 \quad a^2 = b^2 + c^2 - 2cx$$

Pause 1 *choosing the next step*

Only x is left that you cannot measure. What do you replace it with, and where did that replacement come from?

Replace x with $b \cos A$, from the left-hand triangle. That substitution is deliberately saved until last, because x^2 cancels on the line above and doing it earlier would drag a squared cosine through the working for nothing:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Read it as Pythagoras with a correction term. If the angle A opens wider, the side facing it grows, and $-2bc \cos A$ is the amount by which the triangle misses being right-angled.

Follow one example

In triangle ABC, $b = 7$ cm, $c = 9$ cm and the angle between them is $A = 52^\circ$. Find a .

Two sides and the included angle, with the unknown side opposite that angle — this is the cosine rule, and the letters are already in the right places.

$$a^2 = 7^2 + 9^2 - 2 \times 7 \times 9 \times \cos 52^\circ \quad a^2 = 49 + 81 - 126 \cos 52^\circ = 130 - 77.573\dots = 52.426\dots$$

Pause 2 *validity*

Someone stops here and writes $a = 52.4$ cm. What have they forgotten, and how could a glance at the given lengths have caught it?

The square root. The rule gives a^2 , not a . And 52.4 cm is absurd beside sides of 7 cm and 9 cm — no side of a triangle can exceed the sum of the other two, so a must be under 16 cm.

$$a = \sqrt{52.426\dots} = 7.24 \text{ cm (to 3 s.f.)}$$

The condition that matters

Pause 3 *validity*

Put $A = 90^\circ$ into $a^2 = b^2 + c^2 - 2bc \cos A$. What does the formula become, and what does that tell you about the rule?

$\cos 90^\circ = 0$, so the correction term vanishes and the rule collapses to $a^2 = b^2 + c^2$ — Pythagoras. The cosine rule is the general version, and Pythagoras is the special case where the angle happens to be right. That is a useful check: a formula that did not reduce to Pythagoras at 90° would be wrong.

Pause 4 *validity*

If A is obtuse, the foot of the perpendicular falls **outside** the triangle, beyond A . Does the derivation still hold?

Yes. For an obtuse A , $\cos A$ is negative, so $x = b \cos A$ is negative — which is precisely how the algebra records "the foot is on the other side of A ". Every line of the derivation survives, and the $-2bc \cos A$ term becomes an addition, correctly making a longer.

Try a fresh question

5. In triangle PQR , $PQ = 8$ cm, $PR = 5$ cm and $\angle QPR = 112^\circ$. Find QR to 3 significant figures. Check your answer is sensible against the two given sides.

Remember. $a^2 = b^2 + c^2 - 2bc \cos A$ needs two sides and the angle **between** them, with the unknown side opposite that angle. Take the square root at the end. At $A = 90^\circ$ it reduces to Pythagoras.

Check after you try

1. Replace x with $b \cos A$, which came from $\cos A = x \div b$ in the left-hand right-angled triangle. It is left to the end because x^2 cancels first.
2. The square root — the rule gives a^2 . And 52.4 cm is impossible beside sides of 7 cm and 9 cm.
3. It becomes $a^2 = b^2 + c^2$, Pythagoras. The cosine rule is the general case.
4. Yes. $\cos A$ is negative for an obtuse A , so x is negative, which is exactly how the algebra records a foot lying beyond A .
5. $QR^2 = 8^2 + 5^2 - 2 \times 8 \times 5 \times \cos 112^\circ = 89 + 29.97 = 118.97$, so $QR = 10.9$ cm. Sensible: the angle is obtuse, so QR should be the longest side, and $10.9 > 8 > 5$.

Sheet 7 The cosine rule — finding an angle, including obtuse

15 minute revision · Calculator in degree mode

Your aim. You will learn to rearrange the cosine rule to find an angle from three sides, and read a negative cosine correctly.

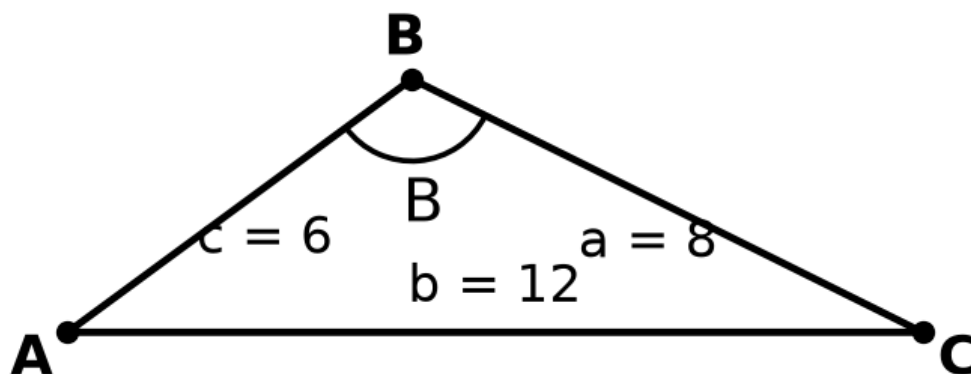
Before you start. $a^2 = b^2 + c^2 - 2bc \cos A$ from Sheet 6. Rearranging an equation to make one letter the subject.

The central idea

Given all three sides and no angle at all, the sine rule is helpless — it has no complete pair to start from. The cosine rule handles it, because it is the one relation connecting three sides to one angle. Make $\cos A$ the subject:

$$a^2 = b^2 + c^2 - 2bc \cos A \quad 2bc \cos A = b^2 + c^2 - a^2$$

$$\cos A = (b^2 + c^2 - a^2) \div (2bc)$$



Triangle ABC with $a = 8$ cm, $b = 12$ cm, $c = 6$ cm. Angle B faces side b . Not to scale.

Note which side is which. The side **a**, alone on the right with the minus sign in front of it, is the side **opposite** the angle you are finding. The other two, b and c , are the sides enclosing it. Getting that the wrong way round is the standard error.

Pause 1 meaning

You are asked for angle B rather than A. Rewrite the formula with the letters in the right places.

$\cos B = (a^2 + c^2 - b^2) \div (2ac)$. The side opposite B is b , so b^2 is the one subtracted; the sides enclosing B are a and c .

Follow one example

In triangle ABC, $a = 8$ cm, $b = 12$ cm and $c = 6$ cm. Find angle B.

Three sides and no angles, so the cosine rule is the method. B is opposite $b = 12$, so b^2 is the subtracted term. Starting from the un rearranged form keeps the letters honest:

$$12^2 = 8^2 + 6^2 - 2 \times 8 \times 6 \times \cos B \quad 144 = 64 + 36 - 96 \cos B \quad 144 = 100 - 96 \cos B \quad 96 \cos B = 100 - 144 = -44 \quad \cos B = -44 \div 96 = -11/24$$

Pause 2 *validity*

The cosine has come out **negative**. Is that a mistake in the working, or is it telling you something?

It is telling you something. Cosine is positive for acute angles and negative for obtuse ones, so a negative value is the algebra announcing that B is obtuse. Keep it exact as $-11/24$ rather than rounding to -0.46 , and take the inverse cosine at the end:

$$B = \cos^{-1}(-11/24) = 117.3^\circ \text{ (to 1 d.p.)}$$

The condition that matters

Pause 3 *choosing the next step*

On Sheet 4 a sine produced **two** candidate angles and you had to test both. Here you have taken $\cos^{-1}$ and stopped at one. Why is there no second candidate to test?

Because cosine behaves differently from sine between 0° and 180° . Sine rises and then falls, so it takes most values twice — once acute, once obtuse. Cosine falls steadily from 1 down to -1 , taking each value exactly **once**, and its sign records which half you are in. So $\cos^{-1}$ returns the only possible answer, and the sign has already settled acute against obtuse for you. **An angle from the cosine rule is never ambiguous.**

Pause 4 *validity*

Check the answer against the shape of the triangle. $B = 117.3^\circ$ is opposite $b = 12$ cm, the longest side. Is that consistent?

Yes — the largest angle always faces the longest side. The other two angles come to 36.3° and 26.4° , and $117.3 + 36.3 + 26.4 = 180.0^\circ$, which is the final check. If an obtuse angle had come out opposite the **shortest** side, the working would be wrong.

Try a fresh question

5. In triangle LMN, $LM = 9$ cm, $MN = 4$ cm and $LN = 7$ cm. Find $\angle LMN$ to 1 decimal place. Then say, without calculating it, which angle of this triangle is the largest, and why.

Remember. $\cos A = (b^2 + c^2 - a^2) \div (2bc)$, where a is the side opposite the angle you want. A negative cosine means an obtuse angle, not an error. There is never a second candidate. Check that the largest angle faces the longest side.

Check after you try

1. $\cos B = (a^2 + c^2 - b^2) \div (2ac)$.
2. Not a mistake — a negative cosine means the angle is obtuse. $B = \cos^{-1}(-11/24) = 117.3^\circ$.
3. Between 0° and 180° cosine takes every value exactly once, so $\cos^{-1}$ gives the only answer. Sine takes most values twice, which is why Sheet 4 needed two candidates.
4. Yes: 117.3° faces the longest side, 12 cm. The three angles total 180.0° .
5. $\angle LMN$ is opposite $LN = 7$, so $\cos M = (9^2 + 4^2 - 7^2) \div (2 \times 9 \times 4) = 48 \div 72 = 2/3$, giving $\angle LMN = 48.2^\circ$. The largest angle is $\angle LNM$, because it faces the longest side, $LM = 9$ cm.

Sheet 8 Bearings

15 minute revision · Calculator in degree mode, ruler, protractor

Your aim. You will learn to read and write three-figure bearings, draw the diagram correctly, and find a reverse bearing.

Before you start. Angles round a point add to 360° . The cosine rule from Sheet 6, for the calculation at the end.

The central idea

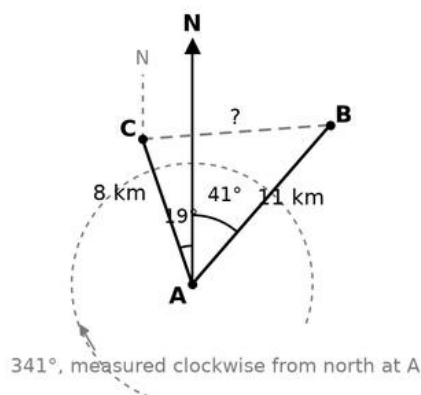
A bearing is a direction, written as an angle. Three rules fix it completely, and almost every lost mark comes from breaking one of them.

Measure from north. Not from the horizontal, not from the nearest road. North is straight up the page, drawn as a line with an arrow and the letter N.

Measure clockwise. Always clockwise, even when anticlockwise looks shorter.

Use three figures. A bearing of forty-one degrees is written **041°**, not 41° . The leading zero is not decoration; it marks the number as a bearing.

There is a fourth rule that is really about drawing rather than arithmetic, and it is the one the notes star: **the north line is drawn at the point you are measuring from.** "The bearing of C from A" is an instruction to stand at A, face north, and turn clockwise until you are looking at C. Draw the north line at A. If you draw it at C you will answer a different question.



From A: C lies 19° west of north, B lies on a bearing of 041° . Not to scale.

Pause 1 meaning

For "the bearing of C from A", at which of the two points do you draw the north line?

At A — the point named after the word *from*, the point you are standing at.

Follow one example

From point A, the point C is 8 km away and lies 19° west of north. The point B is 11 km away on a bearing of 041° . Find (a) the bearing of C from A, (b) the bearing of A from C, and (c) the

distance BC.

(a) Standing at A and turning clockwise from north, you pass through east, south and west before reaching C. That is a full turn less the 19° you would have turned anticlockwise:

bearing of C from A = $360^\circ - 19^\circ = \mathbf{341^\circ}$

Pause 2 *validity*

Why is the answer 341° and not simply 19° ?

Because 19° is measured the wrong way. Turning 19° clockwise from north would take you to the **east** of north, to a completely different point. Bearings are always clockwise, so a direction west of north has a large bearing, over 180° .

(b) The bearing of A from C reverses the journey. Now you stand at C and draw a **new** north line there. The two north lines are parallel, and the reverse direction differs from the original by half a turn:

bearing of A from C = $341^\circ - 180^\circ = \mathbf{161^\circ}$

Subtract 180° when the original bearing is 180° or more; add 180° when it is less. Either way the answer must land between 000° and 360° .

(c) The angle at A between the two paths is the 19° on one side of north plus the 41° on the other, so $\angle CAB = 60^\circ$. That gives two sides and the included angle, which is Sheet 6:

$BC^2 = 8^2 + 11^2 - 2 \times 8 \times 11 \times \cos 60^\circ = 185 - 88 = 97$, so $BC = \mathbf{9.85 \text{ km}}$ (3 s.f.)

The condition that matters

Pause 3 *choosing the next step*

The bearing of B from A is 041° . What is the bearing of A from B?

041° is less than 180° , so add: $041^\circ + 180^\circ = \mathbf{221^\circ}$.

Pause 4 *validity*

A student finds the reverse of 341° by adding: $341^\circ + 180^\circ = 521^\circ$. Repair their method in one sentence.

A bearing cannot exceed 360° , so 521° is not a direction at all. Subtracting 360° from it recovers 161° , the correct answer, but the cleaner rule is to **add 180° when the bearing is under 180° and subtract 180° when it is over**, which keeps you inside the range every time.

Try a fresh question

5. A ship sails from P on a bearing of 070° for 12 km to Q. It then sails on a bearing of 160° for 9 km to R. (a) Show that $\angle PQR = 90^\circ$. (b) Find the distance PR. (c) Find the bearing of R from P, to the nearest degree.

Remember. From north, clockwise, three figures, and the north line drawn at the point you are measuring from. Reverse a bearing by adding 180° if it is under 180° , subtracting 180° if it is over. Keep every answer between 000° and 360° .

Check after you try

1. At A, the point named after the word *from*.
2. Bearings are measured clockwise, so a direction west of north gives a large bearing; 19° clockwise would point east of north instead.
3. 221° , because 041° is less than 180° so you add.
4. No bearing can exceed 360° . Add 180° only when the bearing is below 180° ; otherwise subtract it. Here $341^\circ - 180^\circ = 161^\circ$.
5. (a) At Q the back-bearing to P is $070^\circ + 180^\circ = 250^\circ$, and the new course is 160° , so the turn is $250^\circ - 160^\circ = 90^\circ$. (b) With the right angle at Q, $PR^2 = 12^2 + 9^2 = 225$, so $PR = 15$ km exactly. (c) $\tan \angle QPR = 9 \div 12$, so $\angle QPR = 36.9^\circ$, and the turn at Q was clockwise, so R lies clockwise of Q as seen from P: bearing $= 070^\circ + 36.9^\circ = 107^\circ$ to the nearest degree.

Sheet 9 3D trigonometry — finding the right triangle inside a solid

15 minute revision · Calculator in degree mode

Your aim. You will learn to pull a single two-dimensional right-angled triangle out of a solid and redraw it on its own before calculating anything.

Before you start. Pythagoras' theorem. A vertical edge of a cuboid is perpendicular to every line in the horizontal base, not just to the edges it touches.

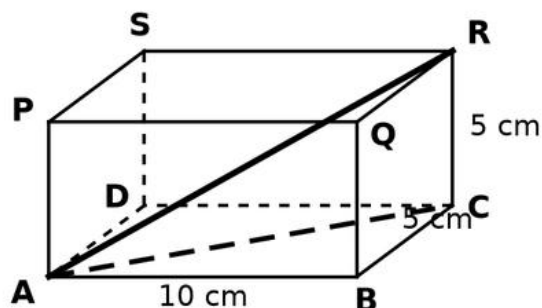
The central idea

Three-dimensional questions are not harder trigonometry. They are ordinary two-dimensional trigonometry with one extra job in front of it: **deciding which triangle to work in**. Everything goes wrong when people try to calculate on the picture of the solid, where the lengths are foreshortened and the right angles do not look like right angles.

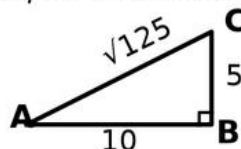
The method is always the same three steps.

Name three points that contain what you know and what you want. **Redraw that triangle on its own**, flat on the page, as large as you like. **Transfer only the lengths you know** onto it, mark the right angle, and forget the solid.

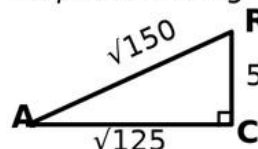
Most cuboid questions need the triangle twice, because the length you want depends on a length nobody gave you — usually a diagonal across the base. So you work along the floor first, then climb.



step 1: in the base



step 2: climbing



Cuboid ABCDPQRS. The base diagonal AC is found first, then the space diagonal AR. Not to scale.

Pause 1 meaning

In triangle ACR, where is the right angle, and how do you know?

At C. The edge CR is vertical, and AC lies flat in the horizontal base, so CR is perpendicular to AC. This is the fact that makes 3D work possible: a vertical edge is at right angles to **every** line in the base, including the sloping diagonals, not only to the edges it visibly meets.

Follow one example

A cuboid ABCDPQRS has a square cross-section 5 cm by 5 cm and length 10 cm, so $AB = 10$ cm, $BC = 5$ cm, and each vertical edge is 5 cm. Find the length of the space diagonal AR.

Pause 2 *choosing the next step*

AR is the hypotenuse of triangle ACR. Which of the two shorter sides do you already know, and which must you find first?

CR = 5 cm is given. AC is not, so AC comes first — and AC is itself the hypotenuse of a right-angled triangle lying flat in the base.

Step 1, in the base. Triangle ABC is right-angled at B, with $AB = 10$ and $BC = 5$:

$$AC^2 = 10^2 + 5^2 = 125, \text{ so } AC = \sqrt{125}$$

Step 2, climbing. Triangle ACR is right-angled at C, with $AC = \sqrt{125}$ and $CR = 5$:

$$AR^2 = (\sqrt{125})^2 + 5^2 = 125 + 25 = 150, \text{ so } AR = \sqrt{150} = \mathbf{12.2 \text{ cm}} \text{ (3 s.f.)}$$

Notice that AC was carried forward as $\sqrt{125}$, not as 11.18. Squaring $\sqrt{125}$ gives exactly 125 and no rounding enters the second step at all.

The condition that matters

Pause 3 *validity*

Suppose you had written $AC = 11.18$ and then $AR = \sqrt{(11.18^2 + 5^2)}$. Would the final answer change? Should you do it that way?

To 3 significant figures it survives here, giving 12.2 cm either way, but it is a habit that fails eventually. Each rounding injects a small error which the next step magnifies. Carry surds or full calculator accuracy and round once, at the end.

Pause 4 *validity*

A student looks at the cuboid, sees a 5 and a 10, writes $AR = \sqrt{(5^2 + 10^2)} = 11.2$ cm, and moves on. Which length have they actually found, and what have they left out?

They have found AC, the diagonal across the **base**, not AR. Their working never uses the vertical 5 cm, so the answer cannot be the space diagonal. A space diagonal involves all three dimensions; if one of the given measurements never appears in the working, the triangle was chosen wrongly.

Try a fresh question

5. A square-based pyramid has base ABCD with $AB = 6$ cm. The apex V lies directly above the centre M of the base, and $VM = 8$ cm. Find (a) the length AM, and (b) the length of the sloping edge VA, each to 3 significant figures. Draw each triangle separately before you calculate.

Remember. Name three points, redraw the triangle flat on the page, and mark the right angle. Work across the base first, then climb. A vertical edge is perpendicular to every line in the base. Carry exact values between steps and round only at the end.

Check after you try

1. At C. CR is vertical and AC lies in the horizontal base, so the two are perpendicular.
2. CR = 5 cm is known; AC must be found first, from triangle ABC in the base.
3. The answer survives to 3 s.f. here, but it is a bad habit — rounding errors accumulate. Carry $\sqrt{125}$ forward.
4. They have found AC, the base diagonal. The vertical 5 cm never enters their working, so it cannot be the space diagonal.
5. (a) AC is the base diagonal, $AC^2 = 6^2 + 6^2 = 72$, and M is its midpoint, so $AM = \sqrt{72} \div 2 = 4.24$ cm. (b) Triangle VMA is right-angled at M, so $VA^2 = AM^2 + VM^2 = 18 + 64 = 82$, giving $VA = 9.06$ cm.

Sheet 10 The angle between a line and a plane

15 minute revision · Calculator in degree mode

Your aim. You will learn to identify the projection of a line onto a plane, and measure the angle between the line and that projection.

Before you start. Sheet 9 — extracting a right-angled triangle from a solid, and finding a base diagonal first. Choosing a trigonometric ratio from Sheet 1.

The central idea

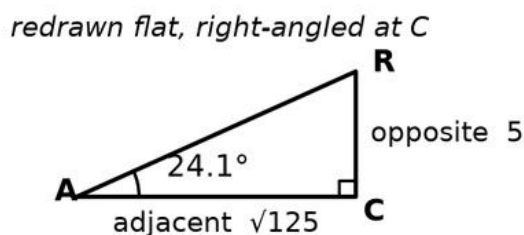
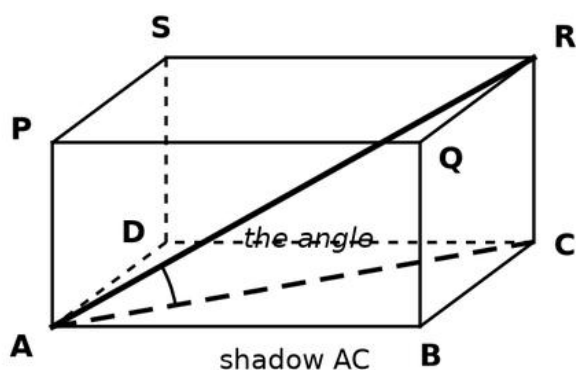
"The angle between the line and the plane" sounds as though you must measure an angle between a stick and a floor, with nothing definite to measure it against. There is a definite answer, and the way to see it is to think about a shadow.

Shine a light straight down on the line. Its shadow on the plane is called the **projection** of the line. The angle you want is the angle between the line and its own shadow.

That gives you three points to work with every time, and one of them is free:

Take the sloping line. Its **top end** is the point up in the air. Its **bottom end** is where it meets the plane, and this point is its own shadow. Then drop a perpendicular from the top end straight down to the plane; call where it lands the **foot**. The shadow runs from the bottom end to the foot.

The triangle made by those three points is always right-angled **at the foot**, because the dropped perpendicular is vertical and the shadow lies flat in the plane. That is the same fact that made Sheet 9 work.



AR and its shadow AC on the base. The angle between them is $\angle RAC$. Not to scale.

Pause 1 meaning

For the diagonal AR in the cuboid above, R is the top end and A is where the line meets the base. Where does R's own shadow land, and so which line on the base is the projection of AR?

R sits directly above C, so C is the foot. The projection of AR is therefore **AC**, the diagonal across the base — and the angle between AR and the base is $\angle RAC$.

Follow one example

The cuboid of Sheet 9 has $AB = 10$ cm, $BC = 5$ cm, and vertical edges of 5 cm. Find the angle between the diagonal AR and the base ABCD.

From Sheet 9, $AC = \sqrt{125} = 11.18$ cm, and $CR = 5$ cm. The angle is $\angle RAC$, in the triangle ACR which is right-angled at C. Redraw it flat before choosing anything.

Now name the angle, $\angle RAC$, and label the sides against it as on Sheet 1. $CR = 5$ is **opposite** it. $AC = \sqrt{125}$ is **adjacent** to it. The hypotenuse AR is not needed.

Opposite with adjacent is the tangent ratio, so tangent is the method.

$\tan(\angle RAC) = 5 \div \sqrt{125} = 0.4472\dots$ $\angle RAC = \tan^{-1}(0.4472\dots) = \mathbf{24.1^\circ}$ (to 1 d.p.)

Pause 2 *choosing the next step*

Sheet 9 also found $AR = \sqrt{150}$. If you had used AR instead of AC, which ratio would you have needed, and would the answer be the same?

Opposite with hypotenuse is sine: $\sin(\angle RAC) = 5 \div \sqrt{150}$, which also gives 24.1° . Either is correct. Tangent is preferable only because it uses the two lengths found directly from the given measurements, so one rounding fewer.

The condition that matters

Pause 3 *validity*

Is $\angle ARC$ the same angle? If not, what does $\angle ARC$ measure?

No. $\angle ARC$ sits at the top of the triangle, between the line AR and the **vertical** edge RC. It measures how far the line leans away from vertical, which is $90^\circ - 24.1^\circ = 65.9^\circ$. The angle asked for is always at the point where the line **meets the plane**, between the line and its shadow.

Pause 4 *validity*

A student uses the edge $AB = 10$ cm as the projection instead of AC, and calculates $\tan^{-1}(5 \div 10) = 26.6^\circ$. Explain why AB is not the projection, and why their answer is too large.

AB is not the shadow of AR, because R does not lie above B — it lies above C. The true shadow AC is longer than AB, since AC is the diagonal and AB only one side of the base. A longer adjacent side with the same opposite side makes the tangent smaller, so the true angle is smaller. Their 26.6° is the angle a different line would make.

Try a fresh question

5. A square-based pyramid has base ABCD with $AB = 6$ cm. The apex V lies directly above the centre M of the base, and $VM = 8$ cm. Find the angle between the sloping edge VA and the base ABCD, to 1 decimal place. State which line is the projection of VA before you calculate.

Remember. The projection is the shadow of the line cast straight down onto the plane. The angle lies at the point where the line meets the plane, between the line and its shadow, in a triangle right-angled at the foot of the perpendicular. Find the shadow's length first.

Check after you try

1. R lies directly above C, so C is the foot and AC is the projection; the angle is $\angle RAC$.
2. Sine, since 5 is opposite and $AR = \sqrt{150}$ is the hypotenuse. It gives the same 24.1° .
3. No. $\angle ARC$ is between AR and the vertical RC, and equals 65.9° . The required angle is at A, where the line meets the base.
4. R is above C, not above B, so AB is not the shadow. AC is longer than AB, which makes the tangent smaller, so the true angle is smaller than their 26.6° .
5. V lies above M, so the projection of VA is AM. From Sheet 9, $AM = 4.243$ cm, and triangle VMA is right-angled at M with $VM = 8$ cm opposite $\angle VAM$. $\tan(\angle VAM) = 8 \div 4.243$, so the angle is 62.1° .

Sheet 11 What makes an argument a proof

15 minute revision · No calculator

Your aim. You will learn to represent an arbitrary integer with a letter, reach a recognisable form, and write the sentence that finishes the proof.

Before you start. An integer is a whole number — positive, negative or zero. You need to expand brackets and collect like terms.

The central idea

A proof is not a demonstration that something works. It is an argument that it *must* work, for every case at once, and the whole trick is one letter.

When you write n and say "let n be any integer", every sentence you then write about n is a sentence about **all** integers simultaneously. That is what lifts an argument above checking examples.

Two representations do most of the work:

an even number is $2n$
an odd number is $2n + 1$

Both say the same thing in words. An even number is a whole number of 2s. An odd number is a whole number of 2s with 1 left over. Consecutive odd numbers are two apart, so they are **$2n + 1$** and **$2n + 3$** — the second is generated from the first by adding 2, not by introducing a new letter.

A proof then ends in a particular way. You manipulate until the expression has a recognisable form, and then you say in words what that form means.

Pause 1 *meaning*

Someone checks that $3 + 5 = 8$ and that $7 + 9 = 16$, and concludes that the sum of two consecutive odd numbers is always even. What is missing from their argument?

They have tested two cases out of infinitely many. Nothing they have written rules out some enormous pair that fails. A proof has to cover every case in one stroke, which is exactly what the letter n does.

Follow one example

Prove that the sum of two consecutive odd numbers is even.

Start by naming what you are talking about, so the reader knows what n means:

Let the two consecutive odd numbers be $2n + 1$ and $2n + 3$, where n is an integer.

$$(2n + 1) + (2n + 3) = 4n + 4 = \mathbf{2(2n + 2)}$$

Pause 2 *choosing the next step*

You have reached $4n + 4$ and rewritten it as $2(2n + 2)$. Why is that particular form the one to aim for?

Because "even" *means* two times an integer. The form $2(2n + 2)$ shows the number is 2 multiplied by something, and $2n + 2$ is an integer because n is. So the last line is not the algebra — it is the sentence:

Since n is an integer, $2n + 2$ is an integer, so the sum is $2 \times$ an integer and is therefore even.

The condition that matters

Pause 3 *validity*

The original notes insert the word "positive" into this statement. Does the proof actually need n to be positive? Try $n = -3$.

It does not. With $n = -3$ the two numbers are -5 and -3 , and their sum is $-8 = 2(-4)$, which is even. Nothing in the algebra ever assumed n was positive. Adding "positive" narrows what is being claimed; it does not repair anything.

Pause 4 *validity*

A student represents the two numbers as $2n + 1$ and $2n + 2$. What have they actually proved?

Not the statement. $2n + 2$ is even, so they have taken one odd number and the next **even** one. Their sum, $4n + 3$, is odd — and the working would appear to disprove a true statement. Choosing the representation is part of the proof, and a wrong one quietly changes the question.

Try a fresh question

5. Prove that the sum of any three consecutive integers is a multiple of 3. Define your letter, show the working, and finish with a sentence.

Remember. Say what your letter stands for, choose a representation that matches the words exactly, manipulate until the form is recognisable, and then write the sentence that says what the form means. Examples illustrate; only the letter proves.

Check after you try

1. They have checked two cases out of infinitely many, which leaves every untested case open. A proof must cover them all at once.
2. Because "even" means $2 \times$ an integer. Reaching $2(2n + 2)$ shows exactly that.
3. No. With $n = -3$ the numbers are -5 and -3 and the sum is -8 , still even. The algebra never used positivity.
4. They have proved something about an odd number and the next even number. Their sum $4n + 3$ is odd, so the wrong representation changed the question.
5. Let the three consecutive integers be n , $n + 1$ and $n + 2$, where n is an integer. Their sum is $n + (n + 1) + (n + 2) = 3n + 3 = 3(n + 1)$. Since n is an integer, $n + 1$ is an integer, so the sum is $3 \times$ an integer and is therefore a multiple of 3.

Sheet 12 One letter, or two?

15 minute revision · No calculator

Your aim. You will learn to decide when two quantities need two different letters, and see what goes wrong when they share one.

Before you start. Sheet 11 — an odd number is $2n + 1$, and a proof ends by naming the form reached. You need to expand two brackets.

The central idea

A letter is a promise about what may vary. Write $2n + 1$ twice and you have written the *same number* twice, because n cannot hold two values at once. So the question to ask before any algebra is: must these two quantities be equal?

If they may differ, they need **different letters**. This is the pointer the original notes star, and it is the difference between proving a statement and proving a special case of it.

Pause 1 *meaning*

Suppose you expand $(2n + 1)(2n + 1)$ and show the result is odd. Which statement have you proved?

That the **square** of an odd number is odd. That is a true and useful statement, but it is not the claim that the product of *any* two odd numbers is odd — it covers 5×5 and says nothing about 5×7 .

Follow one example

Prove that the product of two odd numbers is odd.

The two numbers are not said to be equal, so they get separate letters:

Let the two odd numbers be $2n + 1$ and $2k + 1$, where n and k are integers.

$$(2n + 1)(2k + 1) = 4nk + 2n + 2k + 1$$

Pause 2 *choosing the next step*

You are aiming for the form that means "odd". What is that form, and how do you get $4nk + 2n + 2k + 1$ into it?

Odd means $2 \times$ an integer, plus 1. Take a factor of 2 out of everything except the final 1:

$$= 2(2nk + n + k) + 1$$

Since n and k are integers, $2nk + n + k$ is an integer — products and sums of integers are integers. So the product is $2 \times \text{an integer} + 1$, and is therefore odd.

The condition that matters

Pause 3 *validity*

Point at the exact moment the argument would collapse if both numbers had been called $2n + 1$. What would the final line prove?

The expansion would give $(2n + 1)^2 = 4n^2 + 4n + 1 = 2(2n^2 + 2n) + 1$, which is a correct piece of algebra proving a correct statement — just not this one. Nothing looks wrong on the page. That is what makes the error dangerous: it does not announce itself, it silently shrinks the claim.

Pause 4 *choosing the next step*

Prove that the sum of two odd numbers is even. How many letters do you need, and why?

Two, for the same reason: the two odd numbers need not be equal. With $2n + 1$ and $2k + 1$ the sum is $2n + 2k + 2 = 2(n + k + 1)$, which is $2 \times \text{an integer}$, so it is even.

Try a fresh question

5. Prove that the product of any two consecutive integers is even. You will need to think about which of the two is even — and you may not know which one it is.

Remember. Give different letters to quantities that are allowed to differ. One letter used twice proves only the case where they are equal. When you cannot tell which of two cases holds, prove both.

Check after you try

1. That the square of an odd number is odd — the special case where the two numbers are equal.
2. The form is $2 \times (\text{integer}) + 1$. Take out a factor of 2 from every term but the last: $2(2nk + n + k) + 1$.
3. It would expand to $(2n + 1)^2 = 2(2n^2 + 2n) + 1$ — correct algebra proving that the square of an odd number is odd, not the stated claim.
4. Two letters. $(2n + 1) + (2k + 1) = 2n + 2k + 2 = 2(n + k + 1)$, which is $2 \times$ an integer, so even.
5. Let the two consecutive integers be n and $n + 1$, where n is an integer. One of them is even, but which one depends on n , so take both cases. If n is even, write $n = 2m$: the product is $2m(2m + 1) = 2[m(2m + 1)]$, which is $2 \times$ an integer. If n is odd, write $n = 2m + 1$: the product is $(2m + 1)(2m + 2) = 2[(2m + 1)(m + 1)]$, again $2 \times$ an integer. In both cases the product is even, so it is always even.

Sheet 13 Difference of two squares as a proof tool

15 minute revision · No calculator

Your aim. You will learn to prove a divisibility result by subtracting two squares, and keep $a^2 - b^2$ clearly apart from $(a - b)^2$.

Before you start. Sheets 11 and 12. Expanding $(2n + 3)^2$, and the factorisation $a^2 - b^2 = (a + b)(a - b)$.

The central idea

Many divisibility proofs are a subtraction of two squares, and there are two routes through them. Expanding both squares and subtracting always works. Factorising first with the difference of two squares is faster, and it often hands you the required factor immediately rather than at the end.

$$a^2 - b^2 = (a + b)(a - b)$$

Read the right-hand side: one bracket is the **sum**, the other is the **difference**. Neither is the square of anything.

Follow one example

Prove that the difference between the squares of two consecutive odd numbers is a multiple of 8.

Let the two consecutive odd numbers be $2n + 1$ and $2n + 3$, where n is an integer. The larger is squared first so the result is positive.

Route one — expand both squares.

$$\begin{aligned} & (2n + 3)^2 - (2n + 1)^2 \\ &= (4n^2 + 12n + 9) - (4n^2 + 4n + 1) \\ &= 8n + 8 = \mathbf{8(n + 1)} \end{aligned}$$

Pause 1 *choosing the next step*

Now do the same thing by factorising first. Write down the sum bracket and the difference bracket, and simplify each.

Route two — difference of two squares. With $a = 2n + 3$ and $b = 2n + 1$:

$$\begin{aligned} (2n + 3)^2 - (2n + 1)^2 &= [(2n + 3) + (2n + 1)] \times [(2n + 3) - (2n + 1)] \\ &= (4n + 4)(2) = \mathbf{8(n + 1)} \end{aligned}$$

The $4n^2$ never appears at all. Since n is an integer, $n + 1$ is an integer, so the difference is $8 \times$ an integer and is therefore a multiple of 8.

The condition that matters

Pause 2 *validity*

Is $(2n + 3)^2 - (2n + 1)^2$ the same as $(2n + 3 - 2n - 1)^2$, which is $2^2 = 4$? Test both with $n = 1$ before you answer.

With $n = 1$ the numbers are 5 and 3. The difference of the squares is $25 - 9 = 16$. The square of the difference is $(5 - 3)^2 = 4$. They are not the same, and they are not even close. **$a^2 - b^2$ and $(a - b)^2$ are different expressions:** the first factorises into a sum times a difference, the second expands to $a^2 - 2ab + b^2$. Sheets 15 and 16 draw both, which is the clearest way to keep them apart.

Pause 3 *choosing the next step*

Change the question: take two consecutive **integers** instead, n and $n + 1$, and find the difference of their squares. Is it still a multiple of 8?

$(n + 1)^2 - n^2 = (2n + 1)(1) = 2n + 1$, which is odd — not a multiple of 8, or even of 2. The multiple of 8 came from the numbers being odd and two apart, not from squaring as such.

Pause 4 *meaning*

In the worked example, where exactly did the factor 8 come from?

From the two brackets: the sum gave $4n + 4 = 4(n + 1)$ and the difference gave 2, and $4 \times 2 = 8$. Seeing that is the advantage of the factorising route.

Try a fresh question

5. Prove that the difference between the squares of **any** two odd numbers is a multiple of 8. The two numbers are not consecutive this time, so think about Sheet 12 first. You may use the fact that the product of two consecutive integers is even.

Remember. $a^2 - b^2 = (a + b)(a - b)$ — a sum times a difference, never the square of a difference. Factorising before expanding usually exposes the required factor at once.

Check after you try

1. Sum bracket $(2n + 3) + (2n + 1) = 4n + 4$; difference bracket $(2n + 3) - (2n + 1) = 2$; product $8(n + 1)$.
2. No. With $n = 1$ the difference of the squares is 16 and the square of the difference is 4. The two expressions are different.
3. $(n + 1)^2 - n^2 = 2n + 1$, which is odd. It is not a multiple of 8.
4. From $4(n + 1) \times 2$: the sum bracket supplies a factor 4 and the difference bracket supplies a factor 2.
5. Let the two odd numbers be $2n + 1$ and $2k + 1$, with n and k integers — two letters, because the numbers need not be equal. Then $(2n + 1)^2 - (2k + 1)^2 = (4n^2 + 4n + 1) - (4k^2 + 4k + 1) = 4n^2 + 4n - 4k^2 - 4k = 4[n(n + 1) - k(k + 1)]$. Now $n(n + 1)$ and $k(k + 1)$ are each a product of two consecutive integers, so each is even, and the difference of two even numbers is even. Write that difference as $2m$ with m an integer. The expression becomes $4 \times 2m = 8m$, which is $8 \times$ an integer, so it is a multiple of 8.

Sheet 14 Place value — two-digit numbers and their reverse

15 minute revision · No calculator

Your aim. You will learn to write a two-digit number algebraically, reverse its digits, and prove a result that does not depend on which digits you started with.

Before you start. Place value — in the number 72 the 7 is worth 70. A digit is one of 0 to 9, and the leading digit of a two-digit number cannot be 0.

The central idea

The number written 72 is not 7 times 2. It is 7 tens and 2 units. To do algebra with digits you must write that out:

a two-digit number is $10x + y$

its reverse is $10y + x$

Here x is the tens digit and y is the units digit, each one of the digits 0 to 9, and $x \neq 0$ so that the number really has two digits.

There is a second idea, and it is the one the original notes star. When a result is going to be reversed **again**, you must keep it in place-value form — as a tens part and a units part, with each part a legal digit. Simplifying too early destroys the very structure you are about to use.

Pause 1 *meaning*

Why is the number written $10x + y$ rather than xy ?

Because xy means x multiplied by y . The digits contribute $10x$ and y , so the number is their sum.

Follow one example

Take a two-digit number whose tens digit is larger than its units digit. Subtract its reverse. Add the answer to its own reverse. Show that the result is always 99.

Let the number be $10x + y$ with $x > y$, so the subtraction gives a positive result.

$$(10x + y) - (10y + x) = 10x - 10y + y - x = 10(x - y) + 1(y - x)$$

Pause 2 *choosing the next step*

The units part is $y - x$. Since $x > y$, that is negative — which is not a legal digit. What has to happen before this can be read as a two-digit number?

Borrow ten from the tens column and give it to the units column, exactly as you would in written subtraction. Take 1 away from the tens part and add 10 to the units part:

$$= 10(x - y - 1) + 1(y - x + 10)$$

Now the tens digit is $x - y - 1$ and the units digit is $y - x + 10$, and both sit between 0 and 9. Reverse this number and add:

$$\begin{aligned} & [10(x - y - 1) + (y - x + 10)] + [10(y - x + 10) + (x - y - 1)] \\ &= 11(x - y - 1) + 11(y - x + 10) \\ &= 11[(x - y - 1) + (y - x + 10)] \\ &= 11 \times 9 = \mathbf{99} \end{aligned}$$

Inside the bracket the x and the y cancel: $(x - y - 1) + (y - x + 10) = 9$, whatever the digits were. That is why the answer never changes.

The condition that matters

Pause 3 *validity*

Try it with 32. The subtraction gives $32 - 23 = 9$. What is "the reverse of 9"?

This is the case people lose. 9 is a one-digit number, but here it stands for the two-digit result **09**, with tens digit 0 and units digit 9. Its reverse is 90, and $09 + 90 = 99$ as promised. The algebra knew this all along: with $x = 3$ and $y = 2$ the tens digit is $x - y - 1 = 0$ and the units digit is $y - x + 10 = 9$. The leading zero must be kept.

Pause 4 *validity*

What happens if $x = y$ — say the number is 44?

The subtraction gives 0, there is no two-digit number left to reverse, and the argument does not apply. That is exactly why the question states $x > y$.

Try a fresh question

5. Prove that a two-digit number minus its reverse is always a multiple of 9.

Remember. Write digits out by place value, $10x + y$, and say what x and y may be. If you are going to reverse a result, keep it as a tens part and a units part with both digits legal — borrowing ten where you must — and do not forget a leading zero.

Check after you try

1. Because xy would mean x times y . The tens digit is worth $10x$ and the units digit is worth y .
2. Ten must be borrowed from the tens column and added to the units column, giving $10(x - y - 1) + 1(y - x + 10)$, so that both digits are legal.
3. 9 stands for 09, with tens digit 0. Its reverse is 90, and $09 + 90 = 99$. The formula agrees: $x = 3$, $y = 2$ gives tens digit 0 and units digit 9.
4. The difference is 0, so there is no two-digit number to reverse and the argument does not apply. Hence the condition $x > y$.
5. Let the number be $10x + y$ and its reverse $10y + x$, where x and y are digits. Then $(10x + y) - (10y + x) = 9x - 9y = 9(x - y)$. Since x and y are integers, $x - y$ is an integer, so the difference is $9 \times$ an integer and is therefore a multiple of 9.

Sheet 15 Visual proof — dissecting a square

15 minute revision · Pencil and ruler

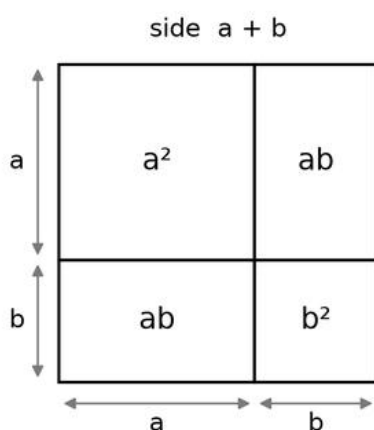
Your aim. You will learn to prove an algebraic identity by computing one area in two different ways, and state the assumptions the picture needs.

Before you start. The area of a rectangle is length $\times$ width. $(a + b)^2$ means $(a + b)(a + b)$, not $a^2 + b^2$.

The central idea

A visual proof rests on one honest observation: **if you cut a shape up and do not throw anything away, the pieces add up to what you started with.** Compute the same area twice, once as a whole and once as a sum of parts, and the two expressions must be equal. That equality is the identity.

Three things make it a proof rather than a picture. Every edge must be labelled, so that each region's area can be read off. Every region must be accounted for, with none left over. And the lengths must be legal — a side of a square cannot be negative or zero, so the picture quietly assumes something about a and b that you should say out loud.



A square of side $(a + b)$, cut once across and once down at a distance a from the corner.

Pause 1 *meaning*

Two of the four regions have area ab . Which are they, and why must they be equal even though one is short and wide and the other is tall and narrow?

The top-right region measures b across by a down; the bottom-left measures a across by b down. Both have area ab , because multiplication does not care about the order.

Follow one example

Prove that $(a + b)^2 = a^2 + 2ab + b^2$.

Each side of the big square is $a + b$, because the cut divides it into a piece of length a and a piece of length b .

The whole. The square has side $a + b$, so its area is $(a + b)^2$.

The parts. Reading the four regions: $a \times a = a^2$, then $b \times a = ab$, then $a \times b = ab$, then $b \times b = b^2$. Their total is $a^2 + 2ab + b^2$.

Pause 2 *choosing the next step*

You now have two expressions for one area. What is the single sentence that turns them into a proof?

Nothing was added and nothing was discarded, so the whole equals the sum of the parts:

$$(a + b)^2 = a^2 + 2ab + b^2$$

The $2ab$ is not an accident of the algebra — it is two genuinely different rectangles that happen to have the same area. That is why the middle term is doubled, which the picture explains better than the expansion does.

The condition that matters

Pause 3 *validity*

What has to be true of a and b for this picture to exist at all?

Both must be positive. A length of zero collapses a region and a negative length has no meaning in a drawing. Note that the identity itself is true for every a and b — it is the **picture** that needs $a > 0$ and $b > 0$, not the algebra.

Pause 4 *choosing the next step*

The next picture proves $(a - b)^2$ instead. Look at the diagram below. What must now be true of a and b , and why?



Try a fresh question

5. Complete the second diagram. Write down the area of each of the four regions A, B, C and D in terms of a and b , add them to get the area of the whole square, and rearrange your equation to prove that $(a - b)^2 = a^2 - 2ab + b^2$. State the assumption you needed.

Remember. Label every edge, account for every region, and write the sentence that says the whole equals the sum of the parts. Say what the lengths must satisfy for the picture to be legal — and remember that is a condition on the diagram, not on the identity.

Check after you try

1. The top-right region is b across by a down, and the bottom-left is a across by b down. Both have area ab .
2. That nothing was added or thrown away, so the area of the whole equals the total of the parts.
3. Both a and b must be positive, or a region collapses or has no meaning. The identity itself holds for all a and b .
4. $a > b$, because the length $a - b$ is drawn in the picture and a length cannot be negative.
5. $A = (a - b)^2$, $B = b(a - b)$, $C = b(a - b)$ and $D = b^2$. The whole square has area a^2 , so $a^2 = (a - b)^2 + 2b(a - b) + b^2$. Rearranging, $(a - b)^2 = a^2 - 2b(a - b) - b^2 = a^2 - 2ab + 2b^2 - b^2 = a^2 - 2ab + b^2$. The assumption needed is $a > b$.

Sheet 16 Visual proof — $a^2 - b^2 = (a + b)(a - b)$

15 minute revision · Pencil and ruler

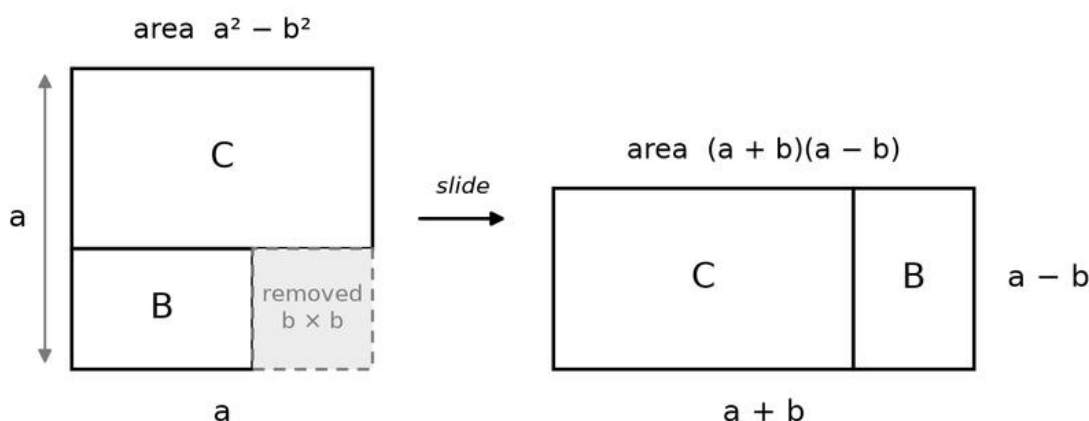
Your aim. You will learn to prove the difference of two squares by cutting a shape up and sliding a piece, and separate the identity from $(a - b)^2$.

Before you start. Sheet 15 — area counted two ways. You should be able to read the area of a rectangle off its labelled edges.

The central idea

Sheet 15 counted one area two ways. This proof does something slightly different: it **moves** a piece. Sliding a piece changes where the area sits but not how much there is, so any two arrangements of the same pieces have equal area — and that equality is the identity.

Start with a square of side a and cut a square of side b out of one corner. What is left is an L-shape, and its area is plainly $a^2 - b^2$: the big square, less the piece removed.



The L-shape cut into two rectangles B and C, then reassembled into a single rectangle.

Pause 1 *meaning*

Look at the two rectangles the L-shape is cut into. One of them measures b by $(a - b)$. Which is it, and what are the dimensions of the other?

B is the bottom-left rectangle, b tall and $a - b$ wide, so its area is $b(a - b)$. C is the top strip, the full width a and $a - b$ tall, so its area is $a(a - b)$.

Follow one example

Prove that $a^2 - b^2 = (a + b)(a - b)$.

The L-shape, before moving anything. Its area is $a^2 - b^2$.

The same two pieces, added up. $a(a - b) + b(a - b)$, which factorises to $(a + b)(a - b)$ because both terms share the factor $(a - b)$.

Nothing has been thrown away, so the two must be equal:

$$a^2 - b^2 = (a + b)(a - b)$$

Pause 2 *choosing the next step*

There is a second way to finish, which needs no factorising. Rotate B and slide it onto the end of C so that the two form a single rectangle. What are the two sides of that rectangle?

Both pieces are $a - b$ in one direction, so they line up along that edge. Laid end to end their other sides, a and b , add to $a + b$. The rectangle is $(a + b)$ by $(a - b)$, so its area is $(a + b)(a - b)$ — and it is made of exactly the pieces whose area was $a^2 - b^2$.

The condition that matters

Pause 3 *validity*

Is $a^2 - b^2$ the same as $(a - b)^2$? Test both with $a = 5$ and $b = 3$ before deciding.

$a^2 - b^2 = 25 - 9 = 16$, while $(a - b)^2 = 2^2 = 4$. They are different, and the two pictures show why: this sheet removes a small square from a big one, whereas Sheet 15 shrinks both sides of the square at once. **$a^2 - b^2$ factorises into a sum times a difference; $(a - b)^2$ expands to $a^2 - 2ab + b^2$.** Nothing turns one into the other.

Pause 4 *validity*

The picture needs $a > b$. Is the identity still true when $a < b$ — say $a = 3$ and $b = 5$?

Yes. $(3 + 5)(3 - 5) = 8 \times (-2) = -16$, and $3^2 - 5^2 = 9 - 25 = -16$. The algebra copes perfectly well with a negative answer; it is only the drawing that cannot, because you cannot cut a square of side 5 out of a square of side 3. As on Sheet 15, the restriction belongs to the picture.

Try a fresh question

5. Use the identity twice. (a) Work out $51^2 - 49^2$ in your head, without squaring either number, and say which two brackets you used. (b) Factorise $9x^2 - 25$ completely.

Remember. $a^2 - b^2$ is a big square with a small square removed, and it rearranges into a rectangle $(a + b)$ by $(a - b)$. It is a sum times a difference. It is not $(a - b)^2$, which is a smaller square.

Check after you try

1. B is the bottom-left rectangle, b by $(a - b)$. The other, C, is the top strip, a by $(a - b)$.
2. The pieces share the side $a - b$, and their other sides a and b add to $a + b$, so the rectangle is $(a + b)$ by $(a - b)$.
3. No: $25 - 9 = 16$ but $(5 - 3)^2 = 4$. One removes a small square from a large one; the other shrinks both sides at once.
4. Yes, the identity holds: $(3 + 5)(3 - 5) = -16$ and $9 - 25 = -16$. Only the diagram needs $a > b$.
5. (a) $51^2 - 49^2 = (51 + 49)(51 - 49) = 100 \times 2 = 200$. (b) $9x^2 - 25 = (3x)^2 - 5^2 = (3x + 5)(3x - 5)$.

Sheet 17 Visual proof — Pythagoras

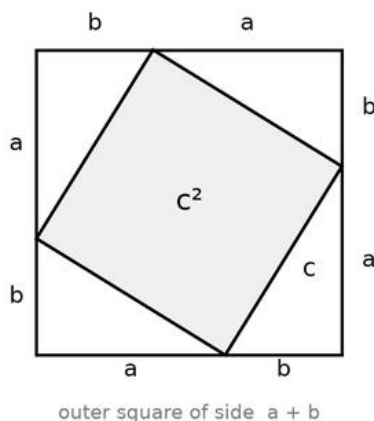
15 minute revision · Pencil and ruler

Your aim. You will learn to prove that $a^2 + b^2 = c^2$ by arranging four identical right-angled triangles inside a square, and justify the step almost everyone skips.

Before you start. Sheets 15 and 16 — area counted two ways. The area of a triangle is $\frac{1}{2} \times \text{base} \times \text{height}$. The two acute angles of a right-angled triangle add to 90° .

The central idea

Take any right-angled triangle with shorter sides a and b and hypotenuse c . Make four identical copies and lay them inside a square of side $(a + b)$, one along each edge, each turned a quarter-turn from the last. A tilted quadrilateral is left in the middle, and every one of its sides is a hypotenuse, so every side has length c .



Four identical right-angled triangles inside a square of side $(a + b)$, leaving a tilted quadrilateral.

Now count the big square's area twice: once from its side, once from its contents.

Pause 1 *meaning*

The four triangles are identical, and each has shorter sides a and b . What is their total area, and why?

Each triangle has area $\frac{1}{2}ab$, taking a as the base and b as the height. Four of them give $4 \times \frac{1}{2}ab = 2ab$.

Follow one example

Prove that $a^2 + b^2 = c^2$.

The whole. The outer square has side $a + b$, so its area is $(a + b)^2$.

The contents. Four triangles, total $2ab$, plus the tilted quadrilateral in the middle, area c^2 .

Nothing overlaps and nothing is left out, so:

$$(a + b)^2 = 4 \times \frac{1}{2}ab + c^2$$

$$a^2 + 2ab + b^2 = 2ab + c^2$$

Pause 2 *choosing the next step*

Both sides now contain $2ab$. What is the next line, and what is left?

Subtract $2ab$ from both sides, which is allowed because you may do the same thing to each side of an equation:

$$a^2 + b^2 = c^2$$

The condition that matters

Pause 3 *validity*

Every side of the middle quadrilateral is c , so it is a rhombus. But the proof called its area c^2 , which is only true of a **square**. Why are its angles right angles?

This is the step that gets skipped, and without it the proof is incomplete. Look at any point where two triangles meet on an edge of the big square. Three angles sit side by side along that straight edge: one acute angle of a triangle, call it α , then an angle of the quadrilateral, then the other triangle's other acute angle, β . Together they make 180° . But α and β are the two acute angles of a right-angled triangle, so $\alpha + \beta = 90^\circ$. That leaves $180^\circ - 90^\circ = 90^\circ$ for the quadrilateral. All four sides equal and all four angles right, so it is a square, and its area really is c^2 .

Pause 4 *validity*

Suppose the four triangles were not right-angled. Would the middle shape still be a square?

No. The argument above used $\alpha + \beta = 90^\circ$ at every corner, which is true only for a right-angled triangle. Without it the middle angles are not 90° , and in fact the four triangles would not fit the corners of a square at all. The right angle is doing real work, which is reassuring — it is what the theorem is about.

Try a fresh question

5. Here is a second arrangement of the same four triangles. Pack them into the corners of a square of side c , with each hypotenuse along an edge. A small square hole is left in the middle, of side $(a - b)$. (a) Write down the area of the big square, and the total area of its contents. (b) Form an equation and simplify it to derive $a^2 + b^2 = c^2$ again.

Remember. Count one area two ways, and account for every piece. Before calling a shape a square, check its angles as well as its sides. Pythagoras comes out of the $2ab$ cancelling from both sides.

Check after you try

1. Each has area $\frac{1}{2}ab$, so four give $2ab$.
2. Subtract $2ab$ from both sides, leaving $a^2 + b^2 = c^2$.
3. At each meeting point on an edge, the two acute angles α and β of the triangles and one angle of the quadrilateral lie on a straight line, so they total 180° . Since $\alpha + \beta = 90^\circ$ in a right-angled triangle, the quadrilateral's angle is 90° . With four equal sides and four right angles it is a square, so its area is c^2 .
4. No. The argument needs $\alpha + \beta = 90^\circ$, which holds only for right-angled triangles; otherwise the middle angles are not right angles.
5. (a) The big square has side c , so its area is c^2 . Its contents are the four triangles, total $4 \times \frac{1}{2}ab = 2ab$, plus the square hole of side $a - b$, area $(a - b)^2$. (b) $c^2 = 2ab + (a - b)^2 = 2ab + a^2 - 2ab + b^2 = a^2 + b^2$.

Sheet 18 Index laws, and zero, negative and fractional indices

15 minute revision · No calculator

Your aim. You will learn to use the three index laws, and see why the zero, negative and fractional indices are forced to mean what they mean.

Before you start. a^n with n a positive whole number means n copies of a multiplied together. Throughout this strand a is positive.

The central idea

Three laws do all the work, and each is just counting factors.

$$a^m \times a^n = a^{m+n} \quad a^m \div a^n = a^{m-n} \quad (a^m)^n = a^{mn}$$

Multiplying puts the two piles of factors together, so you add. Dividing cancels one pile against the other, so you subtract. A power of a power repeats the pile n times, so you multiply.

The other three rules are not extra facts to memorise. They are **forced** by insisting the first law keeps working when the index stops being a positive whole number.

Zero. $a^n \div a^n$ is 1. By the division law it is also $a^{n-n} = a^0$. So $a^0 = 1$.

Negative. $a^0 \div a^n$ is $1 \div a^n$. By the division law it is also $a^{0-n} = a^{-n}$. So $a^{-n} = 1/a^n$: a negative index means a reciprocal, never a negative answer.

Fractional. Raise $a^{1/n}$ to the power n . By the third law that is $a^{n/n} = a$. So $a^{1/n}$ is the number whose n th power is a — the n th root.

Pause 1 *meaning*

Nothing above was proved by checking examples. What single idea forced all three of the new rules?

That the original laws should continue to hold. Once you insist on that, there is no freedom left: each new meaning is the only one that fits.

Follow one example

Evaluate $16^{-3/4}$ without a calculator.

The index carries all three new ideas at once — a minus sign, a denominator and a numerator — so deal with them one at a time.

$$16^{-3/4} = 1 \div 16^{3/4} \text{ (negative index — take the reciprocal)}$$

$$16^{3/4} = (16^{1/4})^3 \text{ (the denominator is a root, the numerator a power)}$$

$$16^{1/4} = 2 \text{ (because } 2^4 = 16)$$

$$\text{so } 16^{3/4} = 2^3 = 8 \text{ and } \mathbf{16^{-3/4} = 1/8}$$

Pause 2 *choosing the next step*

Would you get the same answer by cubing first, working out 16^3 and then taking the fourth root? Which order is easier, and does the order change the answer?

The answer is the same either way, because $(a^m)^n = (a^n)^m$. But cubing first means finding the fourth root of 4096, while rooting first keeps you working with 2. **Take the root first.**

The condition that matters**Pause 3** *choosing the next step*

Write each of $1/9$, 27 and $\sqrt{3}$ as a power of 3.

$1/9 = 3^{-2}$, $27 = 3^3$ and $\sqrt{3} = 3^{1/2}$. Being able to do this instantly is what Sheet 19 depends on.

Pause 4 *validity*

The laws were stated for a positive. Is $(-8)^{1/3}$ allowed? What could go wrong if negative bases were permitted?

The cube root of -8 is -2 , so that one expression looks harmless. But $1/3$ and $2/6$ are the same number, and $(-8)^{2/6}$ would ask for the sixth root of 64, which is $+2$. The same index would give two different answers, so the laws would break. Keeping a > 0 avoids the whole problem, and that is the convention used throughout this strand.

Try a fresh question

5. Evaluate without a calculator, leaving each answer exact: (a) $8^{2/3}$ (b) $25^{-1/2}$ (c) $(16/81)^{3/4}$

Remember. Add the indices to multiply, subtract to divide, multiply for a power of a power. A zero index gives 1, a negative index gives a reciprocal, and a fractional index is a root on the bottom and a power on the top. Take the root first.

Check after you try

1. That the original three laws should keep working. Each new meaning is the only one consistent with them.
2. The same answer, since $(a^m)^n = (a^n)^m$. Taking the root first is far easier: $16^{1/4} = 2$ rather than $16^3 = 4096$.
3. $1/9 = 3^{-2}$, $27 = 3^3$, $\sqrt{3} = 3^{1/2}$.
4. It would break the laws. $(-8)^{1/3} = -2$, but $1/3 = 2/6$ and $(-8)^{2/6}$ would be the sixth root of 64, which is +2. The same index cannot give two answers, so a is kept positive.
5. (a) $8^{2/3} = (8^{1/3})^2 = 2^2 = 4$. (b) $25^{-1/2} = 1 \div \sqrt{25} = 1/5$. (c) $(16/81)^{3/4} = (2/3)^3 = 8/27$, since $16^{1/4} = 2$ and $81^{1/4} = 3$.

Sheet 19 Common base, then equate indices

15 minute revision · No calculator

Your aim. You will learn to turn an equation with different bases into one with a single base, and then read off a linear equation.

Before you start. Sheet 18, especially writing numbers such as 4, 8, $1/9$ and 27 as powers of one base.

The central idea

Almost every exponent question starts with the same move, and it is worth saying out loud before you write anything: **get everything onto one base.**

The reason is a single fact. If a is positive and $a \neq 1$, then the function a^x never repeats a value, so

$$a^m = a^n \implies m = n$$

Once both sides are powers of the same base, the exponents must be equal, and an awkward-looking equation collapses into a linear one.

A small piece of technique goes with it. A constant hiding inside an index can be split off, because adding indices is multiplying:

$$2^{4x+1} = 2^{4x} \times 2^1$$

Pause 1 *meaning*

The rule requires $a \neq 1$. Test it: is it true that $1^2 = 1^5$? What does that tell you about equating indices when the base is 1?

1^2 and 1^5 are both 1, yet $2 \neq 5$. When the base is 1 every index gives the same value, so nothing can be concluded about the exponents. The same failure happens at $a = 0$. That is exactly why the rule carries the conditions $a > 0$ and $a \neq 1$.

Follow one example

Solve $4^{x+1} = 8^{2x-1}$.

The bases 4 and 8 are different, so nothing can be equated yet. But both are powers of 2, and that is the shared base.

$$(2^2)^{x+1} = (2^3)^{2x-1}$$

$$2^{2x+2} = 2^{6x-3}$$

Now both sides are powers of 2, so the indices must match:

Pause 2 *choosing the next step*

Write down the linear equation this gives, and solve it.

$$2x + 2 = 6x - 3$$

$$5 = 4x$$

$$x = 5/4$$

Check by putting it back: $4^{9/4} = 2^{9/2}$, and $8^{3/2} = 2^{9/2}$. They agree.

The condition that matters**Pause 3** *choosing the next step*

Rewrite 2^{4x+1} with the constant split off the index, and then rewrite 3^{2x} as a power of 9.

$2^{4x+1} = 2^{4x} \times 2$. And $3^{2x} = (3^2)^x = 9^x$. Both directions are useful: sometimes you break a base down, sometimes you build it up, and the choice is whichever makes the two sides match.

Pause 4 *validity*

A student meets $5^x = 8$ and writes "5 and 8 are different bases, so there is no common base and no solution." Are they right?

They are right that 5 and 8 share no convenient base — 8 is a power of 2, and 5 is not. They are wrong that there is no solution. A solution exists; it is simply not a fraction, and reaching it needs logarithms, which is Sheet 24. The common-base method is the first thing to try, not the only thing there is.

Try a fresh question

5. Solve $9^{2x-1} = 27^{x+1}$, showing the base you chose and the linear equation you reached. Check your answer.

Remember. Write every term as a power of one base, split any constants off the indices, then equate the indices and solve. The step is valid because $a^m = a^n$ forces $m = n$ whenever a is positive and not 1.

Check after you try

1. Both equal 1, yet the indices 2 and 5 differ. With base 1 the indices cannot be equated at all, which is why the rule excludes $a = 1$.
2. $2x + 2 = 6x - 3$, so $4x = 5$ and $x = 5/4$.
3. $2^{4x+1} = 2^{4x} \times 2$, and $3^{2x} = 9^x$.
4. They are right that there is no shared base, but wrong that there is no solution. The answer is not rational and needs logarithms — see Sheet 24.
5. Both 9 and 27 are powers of 3: $3^{2(2x-1)} = 3^{3(x+1)}$, so $3^{4x-2} = 3^{3x+3}$ and $4x - 2 = 3x + 3$, giving $x = 5$.
5. Check: $9^5 = 3^{18}$ and $27^6 = 3^{18}$.

Sheet 20 When the question wants an expression, not x

15 minute revision · No calculator

Your aim. You will learn to answer a question such as "find 10^x " exactly, without ever finding x.

Before you start. Sheet 19. You also need $a^x \times b^x = (ab)^x$, which is the index laws read the other way round.

The central idea

Some questions do not ask for x at all. They ask for a **compound quantity** such as 10^x or 21^x . It is tempting to find x first and then work the quantity out, but that is the long way round and usually the wrong way: x itself is normally an ugly logarithm, while the quantity asked for is an exact fraction.

The move that makes it work is this law, used from right to left:

$$2^x \times 5^x = (2 \times 5)^x = 10^x$$

So if you can get the equation into a state where 2^x and 5^x sit side by side with nothing else attached, you have 10^x — and you never needed x.

Pause 1 *meaning*

Why is $2^x \times 5^x$ equal to 10^x , and why is $2^x + 5^x$ **not** equal to 10^x ?

Each of 2^x and 5^x is x factors long, so pairing them off gives x copies of 2×5 . Adding gives nothing of the sort — there is no law for a sum of powers. Test it with $x = 1$: $2 \times 5 = 10$, but $2 + 5 = 7$.

Follow one example

Given $2^{4x+1} \times 5^{3x-1} = 2^{3x-2} \times 5^{2x+1}$, find 10^x .

Both bases appear on both sides, so collect each base on one side by dividing. Subtract the right-hand index from the left-hand index, base by base:

$$2^{4x+1 - (3x-2)} \times 5^{3x-1 - (2x+1)} = 1$$

$$2^{x+3} \times 5^{x-2} = 1$$

Pause 2 *choosing the next step*

You want 2^x and 5^x on their own. Split the constants off each index and move them to the other side.

$$2^x \times 2^3 \times 5^x \times 5^{-2} = 1$$

$$2^x \times 5^x = 5^2 \div 2^3$$

$$\mathbf{10^x = 25/8}$$

The whole question is answered and x was never mentioned. For the record x is about -1.29 , an unpleasant decimal — which is precisely why the question did not ask for it.

The condition that matters

Pause 3 *validity*

Would it be wrong to give the answer as 3.125?

It is not wrong, but $25/8$ is better. It is exact, it is shorter, and if the value is used in a later part of the question no rounding has crept in. Keep fractions as fractions unless a decimal is asked for.

Pause 4 *choosing the next step*

Suppose the same question had asked for 20^x instead. Could you answer it from the same information?

Not from one equation. $20^x = 2^{2x} \times 5^x = 2^x \times (2^x \times 5^x)$, so you would need the value of 2^x **on its own** as well as the product. A single equation pins down one combination of 2^x and 5^x , not both separately. Two equations would be needed — which is Sheet 22.

Try a fresh question

5. Given $3^{2x+1} \times 7^{3x-1} = 3^{x-2} \times 7^{2x+1}$, find the exact value of 21^x . Do not find x .

Remember. Collect each base onto one side, split the constants off the indices, and read the remaining product as a single power. $a^x b^x = (ab)^x$ is the law doing the work. Leave the answer as an exact fraction.

Check after you try

1. Because $2^x \times 5^x$ pairs x twos with x fives, giving x copies of 10. Addition has no such law: with $x = 1$, $2 + 5 = 7$ but $10^1 = 10$.
2. $2^x \times 8 \times 5^x \div 25 = 1$, so $2^x \times 5^x = 25/8$, that is $10^x = 25/8$.
3. Not wrong, but $25/8$ is exact and shorter, and avoids rounding if the value is used again.
4. No. $20^x = 2^x \times (2^x \times 5^x)$, so you would also need 2^x by itself. One equation fixes only one combination.
5. Collecting each base: $3^{2x+1-(x-2)} \times 7^{3x-1-(2x+1)} = 1$, so $3^{x+3} \times 7^{x-2} = 1$. Splitting the constants, $3^x \times 27 \times 7^x \div 49 = 1$, so $3^x \times 7^x = 49/27$ and $21^x = 49/27$.

Sheet 21 Quadratics in disguise

15 minute revision · No calculator

Your aim. You will learn to recognise a quadratic hidden inside an exponential equation, substitute to reveal it, and know which roots to throw away.

Before you start. Sheets 18 and 19. You need to factorise a quadratic such as $9y^2 - 10y + 1$.

The central idea

Look for a term in a^{2x} sitting beside a term in a^x . Because $a^{2x} = (a^x)^2$, the equation is a quadratic wearing a costume, and one substitution takes the costume off:

$$\text{let } y = a^x, \text{ then } a^{2x} = y^2$$

Solve the quadratic in y , then go back and solve $a^x = y$ for x . Two things are easy to forget. The substitution has to be undone — a value of y is not an answer. And **a^x is positive for every x** , so any root that makes y zero or negative must be rejected.

Pause 1 *meaning*

Why is 9^{x+1} equal to $9 \times (3^x)^2$? Take it one step at a time.

$9^{x+1} = 9^x \times 9$ by splitting the constant off the index. Then $9^x = (3^2)^x = 3^{2x} = (3^x)^2$. So the whole thing is $9 \times (3^x)^2$, which is $9y^2$ once $y = 3^x$.

Follow one example

Solve $9^{x+1} + 1 = 10 \times 3^x$.

There is a 3^x on the right, and the work above turns the left into $9(3^x)^2 + 1$. Every term is now in 3^x , so the substitution is available.

$$9y^2 + 1 = 10y \text{ where } y = 3^x$$

$$9y^2 - 10y + 1 = 0$$

$$(9y - 1)(y - 1) = 0$$

$$y = 1/9 \text{ or } y = 1$$

Pause 2 *choosing the next step*

Two values of y . The question asked for x . What is the next line for each?

Undo the substitution:

$$3^x = 1/9 = 3^{-2} \implies \mathbf{x = -2}$$

$$3^x = 1 = 3^0 \implies \mathbf{x = 0}$$

Both roots are positive numbers, so both survive, and both check in the original: at $x = 0$, $9 + 1 = 10$ and $10 \times 1 = 10$; at $x = -2$, $1/9 + 1 = 10/9$ and $10 \times 1/9 = 10/9$.

The condition that matters

Pause 3 *validity*

Now solve $4^x - 2^x - 12 = 0$. One root of the quadratic cannot be used. Which, and why?

$4^x = (2^2)^x = (2^x)^2$, so with $y = 2^x$ the equation is $y^2 - y - 12 = 0$, which factorises to $(y - 4)(y + 3) = 0$, giving $y = 4$ or $y = -3$. **Reject $y = -3$** , because 2^x is positive for every real x and can never equal a negative number. That leaves $2^x = 4$, so $x = 2$.

This is the habit to build: after solving the quadratic, look at each root and ask whether an exponential could produce it.

Pause 4 *validity*

Could a substitution $y = a^x$ ever legitimately give $y = 0$?

No. a^x approaches zero for large negative x but never reaches it, so $y = 0$ is rejected just as a negative root is.

Try a fresh question

5. Solve $4^{x+1} - 9(2^x) + 2 = 0$. Show the substitution, state whether either root has to be rejected, and give both values of x .

Remember. A term in a^{2x} beside a term in a^x means a hidden quadratic. Substitute $y = a^x$, solve, reject any root that is zero or negative, then undo the substitution to reach x .

Check after you try

1. $9^{x+1} = 9^x \times 9$, and $9^x = (3^2)^x = (3^x)^2$, so the expression is $9(3^x)^2$.
2. Replace y by 3^x : $3^x = 1/9 = 3^{-2}$ gives $x = -2$, and $3^x = 1 = 3^0$ gives $x = 0$.
3. With $y = 2^x$, $y^2 - y - 12 = 0$ gives $(y - 4)(y + 3) = 0$, so $y = 4$ or $y = -3$. Reject $y = -3$ because 2^x is always positive. Then $2^x = 4$, so $x = 2$.
4. No. a^x gets arbitrarily close to zero but never equals it, so $y = 0$ is rejected too.
5. $4^{x+1} = 4 \times 4^x = 4(2^x)^2$. With $y = 2^x$ the equation is $4y^2 - 9y + 2 = 0$, which factorises to $(4y - 1)(y - 2) = 0$, so $y = 1/4$ or $y = 2$. Both are positive, so neither is rejected. Then $2^x = 1/4 = 2^{-2}$ gives $x = -2$, and $2^x = 2$ gives $x = 1$.

Sheet 22 Simultaneous exponential equations

15 minute revision · No calculator

Your aim. You will learn to turn a pair of exponential equations into a pair of linear ones, and solve them together.

Before you start. Sheet 19. You also need to solve a pair of linear simultaneous equations by substitution or elimination.

The central idea

Two unknowns need two equations, and that is as true for exponentials as for anything else. The useful observation is that **each equation is reduced on its own**. Take the first, force it onto a single base, equate indices, and you have a linear equation in x and y . Do the same to the second. Now solve the linear pair by whichever method you prefer.

The two equations do not have to share a base. One may reduce in base 2 and the other in base 3; the reduction happens separately, and what comes out of each is an ordinary linear equation with no bases left in it at all.

Pause 1 *meaning*

Why does an equation such as $2^{3x} \times 8^y = 1/4$ turn into a **linear** equation rather than something more complicated?

Because once both sides are powers of 2, the indices must match, and those indices are just linear expressions in x and y . The exponential structure disappears entirely at that step.

Follow one example

Solve simultaneously: $2^{3x} \times 8^y = 1/4$ and $27^x \div 3^{y+2} = 1/9$.

First equation, base 2. $8 = 2^3$, so $8^y = 2^{3y}$; and $1/4 = 2^{-2}$.

$$2^{3x} \times 2^{3y} = 2^{-2}$$

$$2^{3x+3y} = 2^{-2}$$

$$3x + 3y = -2$$

Second equation, base 3. $27 = 3^3$, so $27^x = 3^{3x}$; and $1/9 = 3^{-2}$.

$$3^{3x} \div 3^{y+2} = 3^{-2}$$

$$3^{3x-y-2} = 3^{-2}$$

$$3x - y - 2 = -2, \text{ so } 3x - y = 0$$

Pause 2 *choosing the next step*

You now have $3x + 3y = -2$ and $3x - y = 0$. Which is the quicker route, substitution or elimination, and what does the second equation tell you at a glance?

The second rearranges to $y = 3x$, which is ready to substitute:

$$3x + 3(3x) = -2$$

$$12x = -2$$

$$\mathbf{x = -1/6, \text{ and then } y = -1/2}$$

Check in the original: $2^{3x} \times 8^y = 2^{-1/2} \times 2^{-3/2} = 2^{-2} = 1/4$. Correct.

The condition that matters**Pause 3** *validity*

One equation reduced in base 2 and the other in base 3. Is that a problem?

No. Each reduction is self-contained, and once the indices are equated the bases are gone. What you carry forward is a pair of linear equations, which know nothing about where they came from.

Pause 4 *validity*

Suppose the two linear equations had come out as $3x + 3y = -2$ and $6x + 6y = -4$. What would that mean?

They are the same equation, the second being twice the first, so they describe one line rather than two crossing lines. There would be infinitely many solutions, and the two exponential equations would have been carrying the same information in different clothing.

Try a fresh question

5. Solve simultaneously: $5^{2x} \times 25^y = 125$ and $2^{x-y} = 16$. Give x and y as exact fractions and check both in the originals.

Remember. Reduce each equation separately to a single base, equate indices to get a linear equation, then solve the linear pair. The bases need not match between equations, and they vanish at the equating step.

Check after you try

1. Once both sides are powers of one base, the indices must be equal, and those indices are linear in x and y .
2. Elimination is barely needed: $3x - y = 0$ gives $y = 3x$ directly. Substituting, $12x = -2$, so $x = -1/6$ and $y = -1/2$.
3. No. Each equation is reduced on its own, and after equating indices no base remains.
4. The second is twice the first, so they are one equation, not two. There would be infinitely many solutions.
5. First: $25^y = 5^{2y}$ and $125 = 5^3$, so $5^{2x+2y} = 5^3$ and $2x + 2y = 3$. Second: $16 = 2^4$, so $x - y = 4$, giving $x = y + 4$. Substituting, $2(y + 4) + 2y = 3$, so $4y = -5$ and $y = -5/4$, hence $x = 11/4$. Check: $2x + 2y = 11/2 - 5/2 = 3$, and $x - y = 11/4 + 5/4 = 4$.

Sheet 23 Substituting one index relation into another

15 minute revision · No calculator

Your aim. You will learn to get a simple relation out of one equation, put it into a second, and finish the quadratic that appears.

Before you start. Sheets 19, 21 and 22. This sheet puts all three together, so make sure the substitution $y = a^x$ is comfortable.

The central idea

Sheet 22 had two equations that both reduced neatly. Here the second equation refuses to: it has a sum in it, so no amount of rewriting will turn it into "index = index". The way through is to take what the *first* equation gives you — a simple relation between x and y — and substitute it into the second, which then turns out to be a hidden quadratic.

So the shape of the work is: reduce, substitute, clear the fractions, spot the quadratic, solve, and check **both** originals.

Pause 1 *meaning*

Why can an equation such as $3^y = 4 \times 3^{x-2} - 1$ not be solved by equating indices?

Because the right-hand side is a subtraction, not a single power. Equating indices needs one power on each side; a sum or difference of powers has to be dealt with some other way.

Follow one example

Solve simultaneously: $64 \times 4^y = 16^x$ and $3^y = 4 \times 3^{x-2} - 1$.

First equation. Everything is a power of 4: $64 = 4^3$ and $16 = 4^2$.

$$4^3 \times 4^y = 4^{2x}$$

$$4^{3+y} = 4^{2x}$$

$$3 + y = 2x, \text{ so } y = 2x - 3$$

Second equation. Substitute that relation to remove y :

$$3^{2x-3} = 4 \times 3^{x-2} - 1$$

Pause 2 *choosing the next step*

The two negative constants in the indices are in the way. Split them off, then decide what to multiply the whole equation by to clear them.

$3^{2x-3} = 3^{2x} \div 27$ and $3^{x-2} = 3^x \div 9$, so the equation is

$$3^{2x} \div 27 = (4 \times 3^x) \div 9 - 1$$

Multiply every term by 27, the larger denominator:

$$3^{2x} = 12 \times 3^x - 27$$

Now $3^{2x} = (3^x)^2$, so with $u = 3^x$ it is a quadratic:

$$u^2 - 12u + 27 = 0$$

$$(u - 3)(u - 9) = 0$$

$$u = 3 \text{ or } u = 9$$

Both are positive, so neither is rejected. Undoing the substitution, $3^x = 3$ gives $x = 1$ and $3^x = 9$ gives $x = 2$. Then $y = 2x - 3$ gives $y = -1$ and $y = 1$.

The two solutions are $(x, y) = (1, -1)$ and $(2, 1)$.

The condition that matters

Pause 3 *validity*

Check $(2, 1)$ in **both** original equations, not just one. Why does it have to be both?

$64 \times 4^1 = 256 = 16^2$, so the first holds. And $3^1 = 3$ while $4 \times 3^0 - 1 = 3$, so the second holds. Both are needed because a solution of a simultaneous system must satisfy every equation; the relation $y = 2x - 3$ came from the first equation, so checking only the second would never test it.

Pause 4 *validity*

Here both roots survived. What would have made one of them invalid?

A root with u zero or negative, since $u = 3^x$ is always positive. That rejection is the habit from Sheet 21 and it still applies here.

Try a fresh question

5. Solve simultaneously: $32 \times 2^y = 4^x$ and $2^y = 3 \times 2^{x-3} - 1$. Give both pairs (x, y) and check each pair in both equations.

Remember. Reduce the equation that will reduce, and use the relation it gives to remove one letter from the other. Clear the fractional powers by multiplying through, look for the hidden quadratic, reject any non-positive root, and check every answer in both originals.

Check after you try

1. Because the right-hand side is a difference of two terms, not a single power. Equating indices needs one power on each side.
2. $3^{2x-3} = 3^{2x} \div 27$ and $3^{x-2} = 3^x \div 9$. Multiplying through by 27 clears both, giving $3^{2x} = 12 \times 3^x - 27$.
3. $64 \times 4 = 256 = 16^2$, and $3 = 4 \times 3^0 - 1 = 3$. Both must be checked because the relation $y = 2x - 3$ came from the first equation, so the second alone would not test it.
4. A root that was zero or negative, since $u = 3^x$ is always positive.
5. First equation: $32 = 2^5$ and $4^x = 2^{2x}$, so $2^{5+y} = 2^{2x}$ and $y = 2x - 5$. Substituting into the second: $2^{2x-5} = 3 \times 2^{x-3} - 1$, that is $2^{2x} \div 32 = (3 \times 2^x) \div 8 - 1$. Multiplying by 32 gives $2^{2x} = 12 \times 2^x - 32$. With $u = 2^x$, $u^2 - 12u + 32 = 0$, so $(u - 4)(u - 8) = 0$ and $u = 4$ or $u = 8$, both positive. Then $2^x = 4$ gives $x = 2$ and $y = -1$; $2^x = 8$ gives $x = 3$ and $y = 1$. Check $(2, -1)$: $32 \times 2^{-1} = 16 = 4^2$, and $2^{-1} = 0.5$ while $3 \times 2^{-1} - 1 = 0.5$. Check $(3, 1)$: $32 \times 2 = 64 = 4^3$, and $2^1 = 2$ while $3 \times 2^0 - 1 = 2$.

Sheet 24 When the bases will not match — take logarithms

15 minute revision · Calculator, or leave answers exact

Your aim. You will learn to recognise the case where no common base exists, take logarithms of both sides, and leave the answer in exact form.

Before you start. Sheets 19 and 20. This sheet uses logarithms — if you have not met them yet, read Sheets 25 and 26 first and then come back. All you need from them is that $\log(a^n) = n \log a$.

The central idea

Sheet 19 solved exponential equations by forcing a common base. That works whenever the numbers involved are powers of one another. It fails the moment they are not: $5^x = 8$ has a perfectly good solution, but 5 and 8 share no base, so no amount of rewriting will produce "index = index".

Logarithms are what you reach for then, and the reason they help is a single law:

$$\log(a^n) = n \log a$$

A logarithm pulls an index down to the front, where it becomes an ordinary coefficient you can solve for. Two points of technique matter. **Simplify as far as the index laws will take you first** — logarithms are the last resort, not the first move, because they make the algebra heavier. And it does not matter which base of logarithm you use, provided you use the same one on both sides.

Pause 1 *choosing the next step*

In $16^{x-1} = 6^{4x}$, the base 6 is not a power of 2. But 6 is 2×3 . Rewrite 6^{4x} using that, and say which law you used.

$6^{4x} = (2 \times 3)^{4x} = 2^{4x} \times 3^{4x}$, using $(ab)^n = a^n b^n$. This is worth doing because it exposes a factor of 2^{4x} that the left-hand side can cancel with.

Follow one example

Solve $16^{x-1} = 6^{4x}$, giving x in terms of $\log 2$ and $\log 3$.

First the index laws, as far as they go. $16 = 2^4$, so the left side is $2^{4(x-1)} = 2^{4x-4}$; and the right side is $2^{4x} \times 3^{4x}$.

$$2^{4x-4} = 2^{4x} \times 3^{4x}$$

Pause 2 *choosing the next step*

Both sides contain 2^{4x} . What single operation removes it, and why is that better than taking logarithms of the line as it stands?

Divide both sides by 2^{4x} . That leaves one power on each side and nothing else:

$$2^{-4} = 3^{4x}$$

Taking logarithms now is a short job. Doing it a line earlier would have produced three logarithm terms to disentangle instead of two.

$$\begin{aligned}\log(2^{-4}) &= \log(3^{4x}) \\ -4 \log 2 &= 4x \log 3 \\ \mathbf{x} &= \mathbf{-\log 2 \div \log 3}\end{aligned}$$

The condition that matters

Pause 3 *validity*

Does it matter whether you use log to base 10 or ln throughout?

No, as long as the same base is used on both sides. Changing base multiplies every logarithm by the same constant, and here that constant cancels between the top and bottom of the fraction. The answer $-\log 2 \div \log 3$ has the same value either way, about -0.631 .

Pause 4 *validity*

The answer is negative. Is that believable? Check it roughly against the original equation.

It is. With x negative, $x - 1$ is negative too, so the left side 16^{x-1} is a small positive number, and $4x$ is negative so the right side 6^{4x} is also small and positive. Both sides being small fractions is consistent. Substituting $x \approx -0.631$ gives about 0.0109 on each side.

Try a fresh question

5. Solve $5^{x+2} = 3^{2x}$, giving x exactly in terms of $\log 3$ and $\log 5$. Take logarithms of both sides immediately — here there is no common base to exploit first.

Remember. Try the common base first. If the bases genuinely do not match, take logarithms of both sides and use $\log(a^n) = n \log a$ to bring the indices down. Any base of logarithm will do, as long as it is the same on both sides. Leave the answer exact.

Check after you try

1. $6^{4x} = 2^{4x} \times 3^{4x}$, using $(ab)^n = a^n b^n$.
2. Divide both sides by 2^{4x} , leaving $2^{-4} = 3^{4x}$. Taking logarithms then gives two terms rather than three.
3. No, provided the same base is used throughout. Changing base scales every logarithm equally and the factor cancels in the fraction.
4. Yes. A negative x makes both sides small positive numbers, which is consistent; both come to about 0.0109.
5. Taking logarithms of both sides, $(x + 2) \log 5 = 2x \log 3$. Expanding, $x \log 5 + 2 \log 5 = 2x \log 3$, so $2 \log 5 = 2x \log 3 - x \log 5 = x(2 \log 3 - \log 5)$. Hence $x = 2 \log 5 \div (2 \log 3 - \log 5)$, which is about 5.48.

Sheet 25 What a logarithm means, and exact values

15 minute revision · No calculator

Your aim. You will learn to read a logarithm as a statement about powers, and use that to find exact values without a calculator.

Before you start. Sheet 19 — writing numbers as powers of a common base. Nothing about logarithms is assumed.

The central idea

A logarithm answers one question: **what power do I raise this base to, in order to get this number?** Nothing more.

That makes the definition a translation between two ways of writing the same fact:

$$\log_b a = c \text{ means exactly } b^c = a$$

Read the symbols off the second form and the first stops being mysterious. b is the **base**, the number being raised to a power. c is the **answer** the logarithm gives, and it is a power. a is the number you are trying to reach, and it is called the **argument**.

Three conditions come with it, and each is a consequence of the power form. The base must be positive and must not be 1, because 1 to any power is 1 and could never reach anything else. The argument must be positive, because b^c is positive whatever c is.

Pause 1 *meaning*

Rewrite $\log_4 32 = x$ as a statement about powers. Which number is the base, and which is the argument?

$4^x = 32$. The base is 4 and the argument is 32. Notice that the logarithm x is the thing sitting up in the index.

Follow one example

Find the exact value of $\log_4 32$.

Translate to power form first — that is always the opening move:

$$4^x = 32$$

Pause 2 *choosing the next step*

4 and 32 are not powers of each other. What base are they both powers of, and what does the equation become?

Both are powers of 2, so this is Sheet 19 again:

$$(2^2)^x = 2^5$$

$$2^{2x} = 2^5$$

$$2x = 5$$

$$x = 5/2$$

Check it: $4^{5/2} = (\sqrt{4})^5 = 2^5 = 32$. The answer is a fraction, which is normal — there is no reason a logarithm should be a whole number.

The condition that matters

Pause 3 *meaning*

Write down $\log_4 1$ and $\log_4 4$ without any working. Then say why $\log_4 0$ has no value at all.

$\log_4 1 = 0$, because $4^0 = 1$. $\log_4 4 = 1$, because $4^1 = 4$. These two are worth knowing on sight in any base: $\log_b 1 = 0$ and $\log_b b = 1$. But $\log_4 0$ asks for a power of 4 equal to 0, and there is none — 4^c is always positive, getting small but never reaching zero.

Pause 4 *validity*

Why is a base of 1 not allowed? Think about what $\log_1 5$ and $\log_1 1$ would have to mean.

$\log_1 5$ would ask for a power of 1 equal to 5, and there is none, since 1 to any power is 1. $\log_1 1$ is worse: every power of 1 is 1, so the answer could be any number at all. One question with no answer and one with infinitely many — so base 1 is excluded.

Try a fresh question

5. Find the exact value of each, showing the power form you used: (a) $\log_9 27$ (b) $\log_8(1/4)$ (c) $\log_{1/2} 8$

Remember. $\log_b a = c$ and $b^c = a$ say the same thing. Translate to power form, put both sides on a common base, and equate the indices. The base is positive and not 1; the argument is positive.

Check after you try

1. $4^x = 32$, with base 4 and argument 32. The logarithm is the index.
2. Both are powers of 2, giving $2^{2x} = 2^5$, so $2x = 5$ and $x = 5/2$.
3. $\log_4 1 = 0$ and $\log_4 4 = 1$. $\log_4 0$ has no value because every power of 4 is positive and never equals 0.
4. $\log_1 5$ has no answer, since every power of 1 is 1. $\log_1 1$ has infinitely many answers for the same reason. Neither is acceptable, so base 1 is excluded.
5. (a) $9^x = 27$ gives $3^{2x} = 3^3$, so $x = 3/2$. (b) $8^x = 1/4$ gives $2^{3x} = 2^{-2}$, so $3x = -2$ and $x = -2/3$. (c) $(1/2)^x = 8$ gives $2^{-x} = 2^3$, so $x = -3$.

Sheet 26 The laws — splitting

15 minute revision · No calculator

Your aim. You will learn to split a single logarithm of a product, quotient or power into separate terms, combine them back, and state the conditions that make the split legal.

Before you start. Sheet 25, and the index laws from Sheet 18. The original notes call this "splitting", and that is a good word for it.

The central idea

The logarithm laws are not new facts. They are the index laws, seen from the other side. Because a logarithm *is* an index, and indices add when you multiply, logarithms add when you multiply:

$$\begin{aligned}\log(xy) &= \log x + \log y \\ \log(x \div y) &= \log x - \log y \\ \log(x^n) &= n \log x\end{aligned}$$

Multiplication becomes addition, division becomes subtraction, and a power becomes a multiplier out at the front. Two conditions ride along with them, and they are the part students drop. **Every logarithm in the statement must be to the same base.** And **x and y must be positive**, because Sheet 25 showed that is the only time a logarithm exists.

Pause 1 *meaning*

There is no law for $\log(x + y)$. Test whether $\log(x + y) = \log x + \log y$ by putting $x = y = 1$ in base 10.

$\log(1 + 1) = \log 2 \approx 0.301$, while $\log 1 + \log 1 = 0 + 0 = 0$. They are not equal. **A sum inside a logarithm cannot be split at all** — the laws handle products, quotients and powers, and nothing else.

Follow one example

Split $\log(100x^4 \div y^6)$ into separate terms. The base is 10.

Work outwards, taking one law at a time. The outermost structure is a division:

$$\log(100x^4 \div y^6) = \log(100x^4) - \log(y^6)$$

The first term is a product, and the second is a power:

$$\begin{aligned}&= \log 100 + \log(x^4) - \log(y^6) \\ &= \log 100 + 4 \log x - 6 \log y\end{aligned}$$

Pause 2 *choosing the next step*

One of those three terms is an actual number. Which, and what is it?

$\log 100 = 2$, because the base is 10 and $10^2 = 100$. So the answer is

$$2 + 4 \log x - 6 \log y$$

That 2 depends entirely on the base being 10. In base 2, $\log_2 100$ is not a whole number and the term would simply stay as it is. Never simplify a number out of a logarithm without checking the base.

The condition that matters

Pause 3 *validity*

Put $x = -2$. Does $\log(x^4)$ exist? Does $4 \log x$ exist? What does that tell you about the law $\log(x^n) = n \log x$?

$x^4 = 16$, so $\log(x^4) = \log 16$ exists perfectly well. But $\log(-2)$ does not exist, so $4 \log x$ is meaningless. The two sides of the law are **not** interchangeable for negative x : the left-hand side survives and the right-hand side does not. The law holds only for $x > 0$, and writing $4 \log x$ is a claim that x is positive. This is why exam answers are expected to state such assumptions.

Pause 4 *choosing the next step*

Going the other way now. Rewrite $3 \log a$ as a single logarithm, and rewrite the number 1 as a base-10 logarithm.

$3 \log a = \log(a^3)$, using the power law backwards. And $1 = \log 10$, because $10^1 = 10$ — turning a bare number into a logarithm is the trick that lets you absorb it into the others.

Try a fresh question

5. (a) Write $3 \log a - \frac{1}{2} \log b + 1$ as a single logarithm to base 10. (b) Given that $\log 2 = 0.3010$ and $\log 3 = 0.4771$, find $\log 12$ without a calculator.

Remember. Products split into sums, quotients into differences, and powers come down to the front. Sums inside a logarithm do not split at all. Same base throughout, and every argument positive — writing $n \log x$ asserts that $x > 0$.

Check after you try

1. $\log 2 \approx 0.301$ but $\log 1 + \log 1 = 0$, so they are not equal. There is no law for the logarithm of a sum.
2. $\log 100 = 2$, because the base is 10 and $10^2 = 100$.
3. $\log(x^4) = \log 16$ exists, but $4 \log x$ does not, since $\log(-2)$ is undefined. So $\log(x^n) = n \log x$ requires $x > 0$, and writing it asserts that.
4. $3 \log a = \log(a^3)$, and $1 = \log 10$.
5. (a) $3 \log a - \frac{1}{2} \log b + 1 = \log(a^3) - \log(\sqrt{b}) + \log 10 = \log(10a^3 \div \sqrt{b})$. (b) $12 = 2^2 \times 3$, so $\log 12 = 2 \log 2 + \log 3 = 0.6020 + 0.4771 = 1.0791$.

Sheet 27 Solving log equations, and rejecting roots

15 minute revision · No calculator

Your aim. You will learn to combine logarithms into one, convert to power form, solve, and then test every root against the original equation.

Before you start. Sheets 25 and 26. You need to solve a quadratic by factorising.

The central idea

Solving an equation with logarithms in it is three steps: combine into a single logarithm, translate to power form, solve what is left. The fourth step is the one the original notes single out — **rejecting is required** — and it is not optional tidying.

Here is why it is forced. In $\log_2 x + \log_2(x - 2)$, the first term needs $x > 0$ and the second needs $x > 2$, so the whole expression only exists for $x > 2$. Combine them into $\log_2(x^2 - 2x)$ and that expression exists whenever $x^2 - 2x > 0$, which includes negative x as well. **Combining has quietly widened the domain**, so the equation you end up solving is not quite the equation you were given. Any extra roots it produces are artefacts, and they have to be thrown out by checking against the original.

Pause 1 *meaning*

Why does $\log_2 x + \log_2(x - 2)$ become $\log_2(x^2 - 2x)$?

By the product law: the sum of two logarithms to the same base is the logarithm of the product, and $x(x - 2) = x^2 - 2x$.

Follow one example

Solve $\log_2 x + \log_2(x - 2) = 3$.

Combine.

$$\log_2[x(x - 2)] = 3$$

Pause 2 *choosing the next step*

You now have $\log_2(\text{something}) = 3$. What is the very next line?

Translate to power form, exactly as on Sheet 25 — a logarithm equal to 3 means the base raised to 3:

$$x(x - 2) = 2^3 = 8$$

$$x^2 - 2x - 8 = 0$$

$$(x - 4)(x + 2) = 0$$

$$x = 4 \text{ or } x = -2$$

Now test both in the original equation, not in the combined one.

$$x = 4: \log_2 4 + \log_2 2 = 2 + 1 = 3 \checkmark$$

$$x = -2: \log_2(-2) \text{ does not exist } \times$$

Pause 3 *validity*

Which logarithm in the original is it that kills $x = -2$? Name it precisely.

The first one, $\log_2 x$, becomes $\log_2(-2)$, whose argument is negative. The second term is also undefined at $x = -2$, but naming one failure is enough to reject the root. **The solution is $x = 4$.**

The condition that matters

Pause 4 *validity*

Is "x must be positive" a strong enough test here? Check $x = 1$ against the original.

It is not. $x = 1$ is positive, but the second term becomes $\log_2(-1)$, which does not exist. The real condition is that **every** argument must be positive: $x > 0$ and $x - 2 > 0$ together, which means $x > 2$. Working that out at the start is a good habit — it tells you in advance which roots can possibly survive.

Try a fresh question

5. Solve $\log_2 x + \log_2(x + 2) = 3$. State the condition on x before you start, give both roots of the quadratic, and say which is rejected and why.

Remember. Combine to one logarithm, convert to power form, solve, and then test every root in the **original** equation. Combining widens the domain, so extra roots appear that were never solutions. Every argument must be positive.

Check after you try

1. By the product law, $\log_2 x + \log_2(x - 2) = \log_2[x(x - 2)] = \log_2(x^2 - 2x)$.
2. Convert to power form: $x(x - 2) = 2^3 = 8$.
3. $\log_2 x$ becomes $\log_2(-2)$, whose argument is negative, so the root is rejected. The solution is $x = 4$.
4. No. $x = 1$ is positive but makes the second term $\log_2(-1)$, which does not exist. The condition is $x > 0$ and $x - 2 > 0$ together, that is $x > 2$.
5. Both arguments must be positive: $x > 0$ and $x + 2 > 0$, so the condition is $x > 0$. Combining, $\log_2[x(x + 2)] = 3$, so $x^2 + 2x = 8$ and $x^2 + 2x - 8 = 0$, giving $(x + 4)(x - 2) = 0$ and $x = -4$ or $x = 2$. Reject $x = -4$, because $\log_2(-4)$ does not exist. So $x = 2$, and checking, $\log_2 2 + \log_2 4 = 1 + 2 = 3$.

Sheet 28 Change of base

15 minute revision · No calculator

Your aim. You will learn to convert a logarithm from one base to another, and use it to express one logarithm in terms of another.

Before you start. Sheets 25 and 26, especially $\log(x^n) = n \log x$.

The central idea

Any logarithm can be rewritten in any base you like. The formula is

$$\log_b a = \log_c a \div \log_c b$$

where c is whatever new base you choose. It is worth seeing where it comes from, because the derivation is three lines and makes the formula impossible to misremember.

Let $\log_b a = m$. By the definition on Sheet 25 that means $b^m = a$. Take logarithms to base c of both sides, then bring the index down with the power law:

$$\begin{aligned}\log_c(b^m) &= \log_c a \\ m \log_c b &= \log_c a \\ m &= \log_c a \div \log_c b\end{aligned}$$

So the new base goes on the bottom and the old argument stays on top.

Pause 1 *meaning*

In the formula, which of the three numbers a , b and c ends up in the denominator, and what is it?

$\log_c b$ sits underneath, where b is the **old base** — the one you are converting away from. The argument a stays on top throughout.

Follow one example

Given that $\log_2 a = x$, express $\log_8 a$ in terms of x .

You already know a in base 2, so convert the base-8 logarithm into base 2 — always convert towards the base you have information about.

$$\log_8 a = \log_2 a \div \log_2 8$$

Pause 2 *choosing the next step*

What is $\log_2 8$, and why can you write it down instantly?

$\log_2 8 = 3$, because $2^3 = 8$. So

$$\log_8 a = x \div 3 = \mathbf{x/3}$$

The answer makes sense: 8 is a bigger base than 2, so it takes a smaller power to reach the same number a .

The condition that matters

Pause 3 *choosing the next step*

Still with $\log_2 a = x$, express **$\log_a 2$** in terms of x . This one flips the base and the argument.

$$\log_a 2 = \log_2 2 \div \log_2 a = 1 \div x$$

So $\log_a 2$ and $\log_2 a$ are reciprocals. That is worth remembering in its own right: swapping base and argument inverts the logarithm, $\log_b a = 1 \div \log_a b$.

Pause 4 *choosing the next step*

Now express $\log_4(a^2)$ in terms of x . Two things change at once, so deal with the power first and the base second.

The power law gives $\log_4(a^2) = 2 \log_4 a$. Then change base: $\log_4 a = \log_2 a \div \log_2 4 = x \div 2$. So $\log_4(a^2) = 2 \times x/2 = \mathbf{x}$. The doubling and the halving cancel exactly.

Try a fresh question

5. Given that $\log_3 5 = k$, express $\log_9 45$ in terms of k . You will need change of base and one of the laws from Sheet 26 — notice that $45 = 9 \times 5$.

Remember. $\log_b a = \log_c a \div \log_c b$: new base underneath, argument on top. Convert towards the base you already know something about. Swapping base and argument gives the reciprocal.

Check after you try

1. $\log_c b$, where b is the old base. The argument a stays on top.
2. $\log_2 8 = 3$, since $2^3 = 8$. So $\log_8 a = x/3$.
3. $\log_a 2 = \log_2 2 \div \log_2 a = 1/x$. Base and argument swap to give the reciprocal.
4. $\log_4(a^2) = 2 \log_4 a = 2 \times (x \div \log_2 4) = 2 \times x/2 = x$.
5. Change base to 3: $\log_9 45 = \log_3 45 \div \log_3 9$. Now $\log_3 9 = 2$, and $45 = 9 \times 5$ so $\log_3 45 = \log_3 9 + \log_3 5 = 2 + k$. Hence $\log_9 45 = (2 + k)/2$.

Sheet 29 Equations with mixed bases

15 minute revision · No calculator

Your aim. You will learn to solve an equation whose logarithms are to different bases, by converting everything to one base first.

Before you start. Sheets 27 and 28. The change of base formula $\log_b a = \log_c a \div \log_c b$ is used on every line here.

The central idea

$\log_2 x$ and $\log_4 x$ look similar enough that it is tempting to add them. They cannot be added, any more than 3 metres and 3 seconds can. They are answers to different questions — "what power of 2?" and "what power of 4?" — and until they are expressed in the same base they are not like terms.

So the whole method is: **convert every logarithm to one base, then treat the result as an ordinary equation.** Which base? Choose the smallest one present. Converting 4 to 2 gives a tidy fraction; converting 2 to 4 works equally well but tends to leave uglier numbers.

Pause 1 *meaning*

Why can $\log_2 x$ and $\log_4 x$ not simply be added, even though both are logarithms of the same number x ?

They measure x against different bases. $\log_2 8 = 3$ and $\log_4 8 = 1.5$, so the same x gives different values — they are not two copies of one quantity.

Follow one example

Solve $\log_2 x + \log_4 x = 3$.

Convert the base-4 term to base 2:

$$\log_4 x = \log_2 x \div \log_2 4 = \log_2 x \div 2 = \frac{1}{2} \log_2 x$$

Pause 2 *choosing the next step*

Substitute that into the equation. What do the two terms add up to, and what does the equation become?

Now both terms are in base 2, so they are like terms and can be collected:

$$\log_2 x + \frac{1}{2} \log_2 x = 3$$

$$\left(\frac{3}{2}\right) \log_2 x = 3$$

$$\log_2 x = 2$$

$$\mathbf{x = 2^2 = 4}$$

The last step is the translation from Sheet 25: $\log_2 x = 2$ means $x = 2^2$.

The condition that matters

Pause 3 *validity*

Check $x = 4$ in the original equation. Why is checking not an optional extra here?

$\log_2 4 + \log_4 4 = 2 + 1 = 3$, so it works. Checking matters for the same reason as on Sheet 27: the manipulations can widen the domain, and every argument must be positive. Here $x = 4$ is comfortably positive, but the habit is what protects you when the algebra produces a second root.

Pause 4 *choosing the next step*

A harder equation contains $\log_{\sqrt{5}}(2x + 1)$. Convert $\log_{\sqrt{5}} t$ to base 5. What is $\log_5 \sqrt{5}$, and so how many $\log_5 t$ is it?

$\sqrt{5} = 5^{1/2}$, so $\log_5 \sqrt{5} = \frac{1}{2}$. Therefore

$$\log_{\sqrt{5}} t = \log_5 t \div \frac{1}{2} = \mathbf{2 \log_5 t}$$

A square-root base doubles the logarithm. That is the conversion you need whenever a base is a root, and it is worth recognising on sight.

Try a fresh question

5. Solve $\log_3 x + \log_9 x = 3$. Say which base you converted to, show the conversion, and check your answer in the original equation.

Remember. Logarithms to different bases are not like terms. Convert them all to one base — usually the smallest present — collect, solve, translate back to power form, and check in the original.

Check after you try

1. They measure x against different bases, so they are different quantities: $\log_2 8 = 3$ but $\log_4 8 = 1.5$.
2. $\log_2 x + \frac{1}{2} \log_2 x = (3/2) \log_2 x = 3$, so $\log_2 x = 2$ and $x = 4$.
3. $\log_2 4 + \log_4 4 = 2 + 1 = 3$, so it checks. Checking guards against roots introduced by the manipulation and against negative arguments.
4. $\log_5 \sqrt{5} = \frac{1}{2}$, since $\sqrt{5} = 5^{1/2}$. So $\log_{\sqrt{5}} t = \log_5 t \div \frac{1}{2} = 2 \log_5 t$.
5. Convert to base 3: $\log_9 x = \log_3 x \div \log_3 9 = \frac{1}{2} \log_3 x$. The equation becomes $\log_3 x + \frac{1}{2} \log_3 x = 3$, so $(3/2) \log_3 x = 3$, giving $\log_3 x = 2$ and $x = 9$. Check: $\log_3 9 + \log_9 9 = 2 + 1 = 3$.

Sheet 30 Substitution — quadratics in logs and in e^x

15 minute revision · Calculator for the last part

Your aim. You will learn to spot a quadratic hidden inside a logarithmic or exponential equation, substitute, and know which substitutions may go negative and which may not.

Before you start. Sheets 21, 26 and 27. The last part uses e and $\ln$ — if your course does not include them, do the first part and ask for the alternative question.

The central idea

Sheet 21 met a quadratic hiding inside an exponential equation. The same disguise works with logarithms, and the substitution is the same idea: name the repeated thing and call it u .

The new point, and the whole reason these two cases sit on one sheet, is that **the two substitutions behave differently when you come to reject roots**.

$u = \log_2 x$ may be negative — a logarithm takes any real value

$u = e^{3x}$ may not — an exponential is always positive

So a negative root is perfectly acceptable for a logarithmic substitution and must be thrown away for an exponential one. Applying the rule from Sheet 21 blindly here would lose a correct answer.

Pause 1 *meaning*

Three expressions that look alike: $(\log_2 x)^2$, $\log_2(x^2)$ and $\log_2 x^3$. Say what each one means and simplify the last two.

$(\log_2 x)^2$ is the logarithm, then squared — it does not simplify. $\log_2(x^2)$ is the logarithm of x^2 , which by the power law is $2 \log_2 x$. Likewise $\log_2 x^3 = 3 \log_2 x$. Only the first is genuinely quadratic.

Follow one example

Solve $(\log_2 x)^2 - \log_2 x^3 - 4 = 0$.

Use the power law on the middle term first, so that everything is written in terms of the same quantity:

$$(\log_2 x)^2 - 3 \log_2 x - 4 = 0$$

Now $\log_2 x$ appears squared and once on its own, so let $u = \log_2 x$:

$$u^2 - 3u - 4 = 0$$

$$(u - 4)(u + 1) = 0$$

$$u = 4 \text{ or } u = -1$$

Pause 2 *validity*

One root is negative. On Sheet 21 that meant rejection. Does it here?

No — and this is the point of the sheet. $u = \log_2 x$, and a logarithm can be negative; it simply means x lies between 0 and 1. Both roots stand:

$$\log_2 x = 4 \implies x = 2^4 = \mathbf{16}$$

$$\log_2 x = -1 \implies x = 2^{-1} = \mathbf{1/2}$$

Both are positive, so both are genuine solutions. Checking $x = 1/2$: $(-1)^2 - \log_2(1/8) - 4 = 1 - (-3) - 4 = 0$. ✓

The condition that matters

Pause 3 *validity*

Now the other case. In $2e^{6x} - 3e^{3x} - 5 = 0$, put $u = e^{3x}$. What is e^{6x} in terms of u , and what quadratic do you get?

$e^{6x} = (e^{3x})^2 = u^2$, so the equation is $2u^2 - 3u - 5 = 0$.

Pause 4 *validity*

That quadratic factorises as $(2u - 5)(u + 1) = 0$, giving $u = 5/2$ or $u = -1$. Which root survives, and why is the answer different from Pause 2?

Only $u = 5/2$. Here $u = e^{3x}$, and an exponential is positive for every x , so $u = -1$ is impossible. In Pause 2 the substitution was a logarithm, whose value may be any real number. **The test is not "is the root negative?" but "can the thing I substituted ever take that value?"**

Try a fresh question

5. Solve $2e^{6x} - 3e^{3x} - 5 = 0$, giving x to 3 significant figures. Start from the quadratic you formed in Pause 3, reject the impossible root, and finish with natural logarithms.

Remember. Name the repeated quantity and substitute. Use the power law first so every term is in the same quantity. Then reject by asking what values the substituted expression can actually take — a logarithm may be negative, an exponential may not — and undo the substitution.

Check after you try

1. $(\log_2 x)^2$ is the logarithm squared and does not simplify. $\log_2(x^2) = 2 \log_2 x$, and $\log_2 x^3 = 3 \log_2 x$.
2. No. $u = \log_2 x$ may be negative, which just means $0 < x < 1$. So $\log_2 x = 4$ gives $x = 16$ and $\log_2 x = -1$ gives $x = 1/2$, and both are valid.
3. $e^{6x} = (e^{3x})^2 = u^2$, so the equation becomes $2u^2 - 3u - 5 = 0$.
4. Only $u = 5/2$, because $u = e^{3x}$ is positive for every x , so $u = -1$ cannot occur. In Pause 2 the substitution was a logarithm, which may take negative values. The test is what the substituted expression is capable of, not the sign of the root.
5. From $2u^2 - 3u - 5 = 0$ with $u = e^{3x}$, factorising gives $(2u - 5)(u + 1) = 0$, so $u = 5/2$ or $u = -1$. Reject $u = -1$ since $e^{3x} > 0$ always. Then $e^{3x} = 5/2$, so $3x = \ln 2.5$ and $x = (\ln 2.5) \div 3 = 0.305$ to 3 s.f.